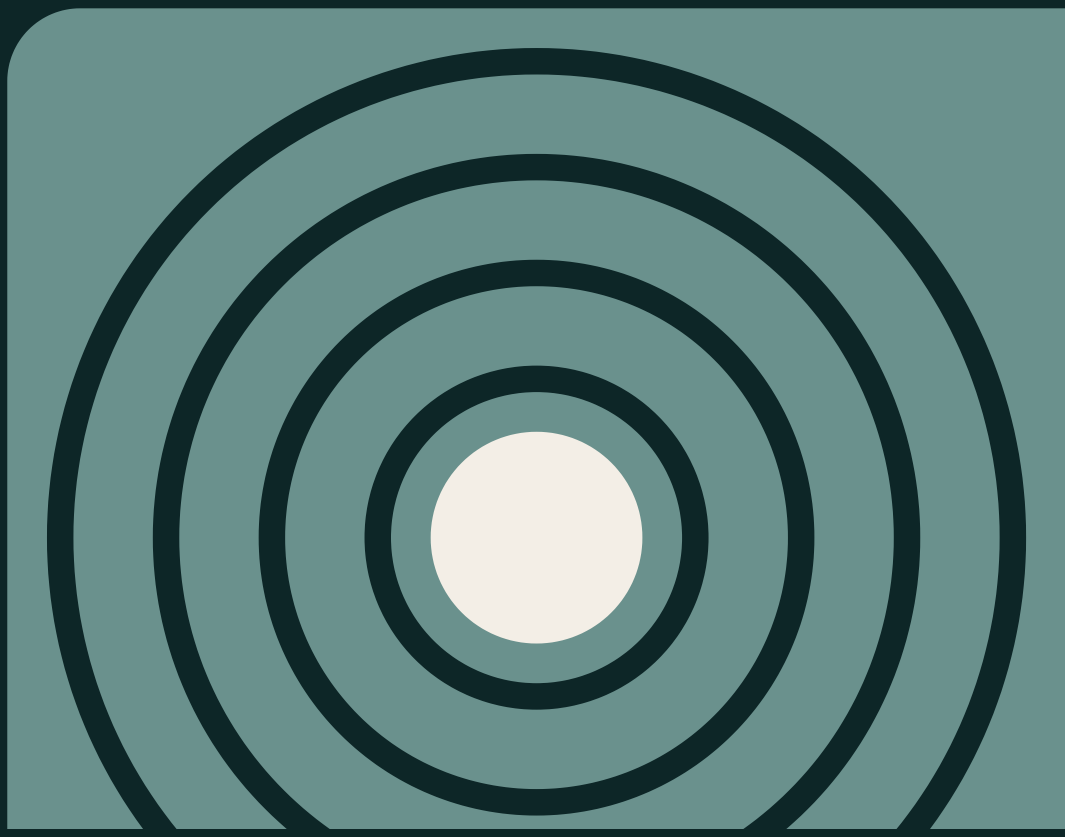


Nacre

The design system for Kushwant KR's portfolio

Version 2, 22 October 2026. Placeholder content throughout; replace it before launch.

Contents

The design system for Kushwant KR's portfolio: a deep lagoon, one iridescent pearl, and type from an Indian foundry.

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Brand book

Nacre is the visual language of Kushwant KR's portfolio: a deep lagoon, one iridescent pearl, and type from an Indian foundry. Hyderabad is the City of Pearls, and a pearl grows layer by layer, the way an engineer does. Keep everything quiet so the pearl stays the one memorable thing.

Principles

1. **The pearl is the only ornament.** The 3D pearl in the hero, the headline's full stop, the KR monogram's dot, and the Pearl chat launcher are one object at four sizes. Nothing else glows, floats, or carries a gradient.
2. **Round means press.** Use `radius-pill` and circles only for things a visitor presses: buttons, chips, the launcher, the send button. Things you read or fill in use `radius-sm` to `radius-lg`.
3. **Layers, not cards.** Content sits on tonal layers: `lagoon`, then `reef`, then `shoal`. Give an element a fill or an outline only when it is a separate object, such as a case study you can open.
4. **Evidence looks like evidence.** Dates, metrics, and links get one treatment everywhere: tabular figures, numbers in `pearl`, links in `rose` with an underline.
5. **Motion answers people.** Two things move unprompted: the hero's opening, where the pearl grows out of the full stop, and the particles responding to scroll. Everything else moves only when a visitor acts.

Colour

- Set the page and the 3D canvas on `lagoon`. Use `reef` for case-study cards, form fields, code blocks, and the chat panel; use `shoal` for hover states, visitor chat bubbles, menus, and toasts.
- Set text in `pearl` and secondary text in `mist`. Both reach 5.2:1 or more on `lagoon`, `reef`, and `shoal` in both themes.
- `rose` is the single accent: primary buttons, links, the active nav underline, list markers, and the focus ring. Use one rose-filled button per view, with `on-rose` text on it.
- `seaglass` means available or success, `amber` means limited or warning, `coral` means error. Pair each with a word or an icon, and put it on its `-soft` ground. Success leans blue-green and danger leans orange-red, so they never rely on a red-green distinction.
- Use `line` for decoration only. The boundary of any control is `line-strong` (3.4:1 or more against every surface).
- The `scene-*` tokens are for WebGL and the CSS pearl only.
- Never use gradients in interface chrome, purple or blue glows, neon green, or pure black or white grounds.
- The dark Lagoon theme is the default and ships at launch. The light Pearl theme is defined for P2: it swaps grounds and text and darkens the accent so every pair still clears 4.5:1.

Pair (Lagoon theme)	Contrast
`pearl` on `lagoon`	13.7:1
`mist` on `reef`	6.7:1
`rose` on `lagoon`	8.4:1
`on-rose` on `rose`	8.4:1
`seaglass` on `seaglass-soft`	6.9:1
`line-strong` on `shoal`	3.5:1

Typography

- Use one family, Anek Latin from Ek Type in Mumbai, variable in weight (100–800) and width (75–125). Width carries the hierarchy: `display` at wdth 114, `h1` 110, `h2` 108, `h3` 104, everything else 100.
- Use the styles by role: `display` for the hero headline only, `h1` for case-study titles, `h2` for section titles, `h3` for sub-headings and featured project titles, `h4` for item titles, `lead` for the sub-headline and section summaries, `body` for running text, `ui` for buttons and labels, `small` for badges and metadata, `caption` for image captions and the chat disclosure.

- Sizes come from the classical scale (12, 14, 16, 18, 21, 24, 36, 48, 60, 72) and are fluid between 360 and 1440 px; each style's usage note gives its `clamp()`.
- Keep running text at `measure` (68ch) or less, line height 1.6. Balance headings with `text-wrap: balance`.
- Write in sentence case everywhere. No all-caps labels, no letterspaced eyebrows, and no single word in a heading set in italic, bold, or a second colour.
- Telugu text falls through the stack to Anek Telugu, the same design family; set Pearl's Telugu answers in `telugu-body` (line height 1.7).
- Use Martian Mono (`code`, `code-inline`) for code and nothing else.
- Use tabular figures for dates, metrics, and timelines.

Shape, borders, and elevation

- Radii by role: `radius-xs` for inline code and keys; `radius-sm` for inputs, badges, and media; `radius-md` for cards, code blocks, and toasts; `radius-lg` for the chat panel, sheets, and the command palette; `radius-pill` for anything pressable.
- Borders are 1 px `line` for decoration and 1.5 px `line-strong` for controls.
- Elevation comes from the surface step plus `shadow-lustre`, a one-pixel highlight along the top edge. Use `shadow-float` only for things that float over the page: the chat panel, the launcher, toasts, the command palette. There are no other shadows.
- Every interactive element shows `shadow-focus` on keyboard focus: a 2 px `lagoon` gap, then a solid 2 px `rose` ring.

Layout

- A 12-column grid inside `container-max` (1200 px), with a side gutter of `max(16px, env(safe-area-inset-left))` on phones, 32 px on tablets, and 48 px on desktop.
- Align everything to the left edge of the grid. Nothing is centred except the 404 page.
- Section padding is `space-10` on desktop, `space-9` on tablets, and `space-8` on phones. The gap from a section header to its content is `space-7`.
- The Layout section has breakpoints and a wireframe of every section.

The pearl, in CSS and in 3D

- In CSS, `.kr-pearl` paints the pearl from `scene-pearl`, `scene-sheen-rose`, `scene-sheen-aqua`, and a white highlight, at `pearl-dot` (0.32em) when it replaces a full stop.
- In WebGL it is a physical material with iridescence and clearcoat under procedural studio light; the Motion and 3D section lists every parameter.
- Use the pearl in exactly four places: the hero (3D), the headline's full stop, the monogram, and Pearl (launcher, avatar, typing dots). Nowhere else.

Iconography

- Use Lucide icons (lucide-react) at 20 px with a 1.75 stroke in `currentColor`, and 16 px inside small buttons and toasts.
- No emoji in the interface, the copy, or Pearl's answers.
- Every icon-only button carries an `aria-label`. The launcher has no icon: the pearl is the icon.
- There is no logo file. The monogram is live type: "KR" in Anek Latin 780 at wdth 112, followed by the pearl dot.

Imagery

- Show real screenshots and screen recordings of the work at 16:10 on a `shoal` mat with `radius-sm`. Videos are muted, looped, under 2 MB, with a poster frame.
- No stock photos, no device-frame mockups, no AI-generated illustrations.
- An optional portrait is duotone: `lagoon` in the shadows, `pearl` in the highlights.

Content fundamentals

- The site speaks in the first person ("I built..."). Pearl speaks about Kushwant in the third person and never pretends to be him.
- Be confident, specific, and warm. Lead with outcomes and numbers, and write numbers in digits.
- Use Indian digit grouping for rupees (₹1,50,000) and plain digits elsewhere.

- Never write: passionate, hard-working, ninja, rockstar, guru, cutting-edge, synergy, or a "Hi, I'm..." greeting with a waving hand.
- Model copy: "I build web products that think and move." "Hire me for a project." "Freelance: 1 slot open in November." "Enter an email address like name@example.com." "Message sent. Kushwant replies within 24 hours."
- The Writing section has the full list of interface copy.

Accessibility

- Meet WCAG 2.2 AA: the contrast pairs above, focus always visible and never hidden under the sticky bar, targets of at least 24 px (primary actions `target-min`, 44 px), and reflow at 320 px with no horizontal scroll.
- The canvas is `aria-hidden` and carries no information of its own; everything it shows also exists in the HTML.
- Honour `prefers-reduced-motion`: no opening move, no morphing, no shimmer. The 3D scene renders one still frame.
- Pearl's log is `role="log"` with polite announcements, and its confirm card is a real form with real buttons.

Using these files in the codebase

- `tokens.json` is the single source. The Implementation section shows the Tailwind v4 theme, the fonts, and a script that turns `tokens.json` into `src/styles/tokens.css`.
- `components/bundle.css` is a plain-CSS reference for every component; `components/index.d.ts` is the React prop contract each component in the codebase implements.

Colour

Every colour in the system, in both themes. Lagoon (default) ships at launch; Pearl (P2) is defined for later.

Token	Lagoon (default)	Pearl (P2)	Use it for
`lagoon`	`#0d2627`	`#f0f3f1`	Page background and the 3D canvas clear colour. Every section sits directly on it.
`lagoon-veil`	`#0d2627db`	`#f0f3f1db`	The sticky navigation bar after 80 px of scroll, with a 12 px backdrop blur. Text on it uses pearl and mist as on lagoon.
`reef`	`#123131`	`#fafbfa`	First layer: case-study cards, media mats, form fields, code blocks, the chat panel body.
`shoal`	`#1a3c3b`	`#e4eae7`	Raised layer: hovered cards, the visitor's chat bubbles, menus, the command palette, toasts.
`line`	`#2e5351`	`#c9d5d1`	Decorative hairlines: card outlines, dividers between list rows, the timeline's growth line. Never the only boundary of a control.
`line-strong`	`#6a918d`	`#6b827d`	Boundaries of controls (inputs, outline buttons, filter chips) and past-role timeline rings: 3.4:1 or more on lagoon, reef and shoal in both themes.
`pearl`	`#f3eee6`	`#0d2627`	Primary text and headings on lagoon, reef and shoal: 10.4:1 or more in both themes.
`mist`	`#a3b9b5`	`#4a6461`	Secondary text (summaries, metadata, captions, placeholders) on lagoon, reef and shoal: 5.2:1 or more in both themes.
`rose`	`#f4a7b9`	`#a83d5a`	The one accent: primary button fill, links, the active nav underline, the focus ring, list markers. As text on lagoon, reef, shoal and rose-soft: 4.7:1 or more in both themes.
`rose-hover`	`#f8c0cc`	`#8f3049`	Hover and pressed state of rose fills and rose links.
`on-rose`	`#0d2627`	`#ffffff`	Label text and icons on a rose or rose-hover fill: 7.8:1 or more. Never literal white in the dark theme.
`rose-soft`	`#2d383b`	`#e9e1e2`	Ground behind rose text: the selected filter chip, the active command-palette row.
`seaglass`	`#86d8c3`	`#1c6e5c`	Available and success: the availability dot, success text and icons. Always paired with a word.
`seaglass-soft`	`#1e3f3d`	`#dbe6e2`	Ground behind seaglass text in status badges and success toasts (4.8:1 or more).
`amber`	`#f1c46b`	`#855a00`	Limited and warning: limited availability, rate-limit notices. Always paired with a word.
`amber-soft`	`#2d3c31`	`#e5e4d9`	Ground behind amber text (4.7:1 or more).
`coral`	`#ff8e7a`	`#b3261e`	Errors: field errors and failed sends. Always with an icon and a sentence that says how to fix it.
`coral-soft`	`#2f3533`	`#eadedc`	Ground behind coral text in error banners and error toasts (5.0:1 or more).
`scene-particle`	`#cfede5`	`#2a5c58`	WebGL only: the main particle colour. Additive blending in the dark theme, normal blending in the light theme.
`scene-particle-glow`	`#86d8c3`	`#1c6e5c`	WebGL only: the brightest 15% of particles, and the pulse while Pearl is answering.
`scene-pearl`	`#ece6ef`	`#f6f2f5`	WebGL only: base colour of the pearl's physical material; also the body of the CSS pearl.
`scene-sheen-rose`	`#f4a7b9`	`#e8a0b2`	WebGL only: the warm iridescent overtone and sheen colour; also the CSS pearl's warm edge.
`scene-sheen-aqua`	`#9ee6e0`	`#8fd3cc`	WebGL only: the cool iridescent overtone and rim light; also the CSS pearl's cool edge.
`scene-fog`	`= lagoon`	`= lagoon`	WebGL only: fog colour, an alias of lagoon so distant particles dissolve into the page in either theme.

Typography

Anek Latin for everything, Anek Telugu for Telugu, Martian Mono for code. Sizes below are the desktop end of each fluid style.

Display and headings

Width carries the hierarchy: set font-variation-settings 'wdth' as each usage says. Sizes are the desktop end of a fluid clamp() between 360 and 1440 px, on the classical scale 12-14-16-18-21-24-36-48-60-72.

`display` 72px / 1.02, weight 760, width 114

I build web products that think and move

The hero headline only, once per site. wdth 114. Fluid clamp(3rem, 2.5rem + 2.222vw, 4.5rem). Its full stop is the pearl dot.

`h1` 60px / 1.05, weight 720, width 110

Clinic booking assistant

Case-study page titles. wdth 110. Fluid clamp(3rem, 2.75rem + 1.111vw, 3.75rem).

`h2` 48px / 1.08, weight 700, width 108

Freelance and client work

Home-page section titles. wdth 108. Fluid clamp(2.25rem, 2rem + 1.111vw, 3rem).

`h3` 36px / 1.15, weight 660, width 104

What I built

Sub-headings inside case studies and featured project titles. wdth 104. Fluid clamp(1.5rem, 1.25rem + 1.111vw, 2.25rem).

`h4` 24px / 1.25, weight 620, width 100

Associate Software Engineer

Item titles: roles, services, card titles, tier names. wdth 100. Fluid clamp(1.3125rem, 1.25rem + 0.2778vw, 1.5rem).

Text

wdth 100 for all running text. Body measure 68ch at most. Weights sit slightly above 400 because light text on lagoon reads thinner than dark text on white.

`lead` 21px / 1.5, weight 460

Full-stack apps with a soft spot for AI and 3D.

The hero sub-headline and the one-line summary under every section title, in mist. Fluid clamp(1.125rem, 1.0625rem + 0.2778vw, 1.3125rem).

`body` 18px / 1.6, weight 460

I turned a phone-only booking process into a website chat that books consultations.

Running text in pearl. Fluid clamp(1rem, 0.9583rem + 0.1852vw, 1.125rem). Max 68ch.

`ui` 16px / 1.25, weight 600

Hire me for a project

Buttons, nav links (at 560), form labels, the chat composer.

`small` 14px / 1.5, weight 520

Freelance: 1 slot open in November

Badges, chips, metadata, form hints, chat source chips, timeline dates (tabular figures).

`caption` 12px / 1.4, weight 540

Booking flow on mobile, week 3 demo

Image captions and the chat disclosure line only, in mist. Never for anything a visitor must act on.

Indic

Pearl answers in the visitor's language. Telugu falls through the sans stack to Anek Telugu, the same design family from the same foundry, so mixed-script answers keep one voice.

`telugu-body` 18px / 1.7, weight 460

నమస్కారం

Telugu text in Pearl's answers. Telugu needs the taller 1.7 line height for its stacked conjuncts.

Code

Code only. Never for labels, metadata, dates or numbers in running text.

`code` 15px / 1.6, weight 420, width 87

```
const pearl = useRef<Mesh>(null)
```

Code blocks in case studies and in Pearl's developer answers. wdth 87.5.

`code-inline` 0.9em / 1.4, weight 460

`useFrame`

Inline code inside running text, on reef with radius-xs.

Space, shape and elevation

Spacing

A 4 px base. Lay out siblings with gap, never per-element margins.

Token	Value		Use it for
space-1	4px		Icon-to-text nudges; the gap inside a badge's dot and label.
space-2	8px		Gap between chips and badges; button icon gap; label-to-field gap.
space-3	12px		Gap between a heading and its summary; badge and chip side padding.
space-4	16px		Minimum page gutter on phones; card padding on phones; gap between form fields.
space-5	24px		Card padding; button side padding; grid column gap; chat panel padding.
space-6	32px		Gap between a card's media and text; nav side padding on desktop.
space-7	48px		Gap between a section header and its content; between timeline entries.
space-8	64px		Section padding on phones; gap between case-study rows.
space-9	96px		Section padding on tablets.
space-10	128px		Section padding on desktop; the breathing room above the hero headline.

Radius

Shape encodes meaning: a pill or circle is something you press; a rounded rectangle is something you read or fill in.

Token	Value		Use it for
radius-xs	4px		Inline code, keyboard keys, focus target of nav links.
radius-sm	8px		Inputs, badges and tags (read-only), media inside cards, tooltips.
radius-md	14px		Case-study cards, code blocks, chat cards, toasts, the chat composer.
radius-lg	22px		The chat panel, the mobile menu sheet, the command palette.
radius-pill	999px		Things you press: buttons, filter and starter chips, source chips, the send button, the pearl launcher.

Shadows

Elevation comes from surface steps first. These three shadows are the only ones in the system.

Token	On lagoon	Value	Use it for
shadow-lustre		`inset 0 1px 0 #f3eee60f` Light: `inset 0 1px 0 #ffffffcc`	A one-pixel top highlight on reef and shoal surfaces, like light on nacre. Cards, chat panel, toasts.
shadow-float		`0 24px 48px #03111166, 0 2px 8px #03111140` Light: `0 18px 40px #0d262724, 0 2px 6px #0d262714`	Only for things that float over the page: the chat panel, the pearl launcher, toasts, the command palette.
shadow-focus		`0 0 0 2px #0d2627, 0 0 0 4px #f4a7b9` Light: `0 0 0 2px #f0f3f1, 0 0 0 4px #a83d5a`	The keyboard focus ring on every interactive element: a 2 px gap in lagoon, then a solid 2 px rose ring (5.4:1 or more on every surface).

Sizes

Fixed dimensions that layout and components share.

Token	Value	Use it for
container-max	1200px	Maximum content width; the 12-column grid lives inside it.

Token	Value	Use it for
`measure`	`68ch`	Maximum width of running text.
`nav-height`	`72px`	Navigation bar height on desktop (64px on phones); also scroll-margin-top for section anchors.
`hero-min-height`	`min(100svh, 920px)`	Hero height: fills a phone screen, never taller than 920px on large monitors.
`chat-panel-width`	`400px`	Pearl's side panel on desktop; phones use a 90svh bottom sheet.
`launcher-size`	`56px`	The pearl launcher's diameter (48px on phones).
`pearl-dot`	`0.32em`	Diameter of the CSS pearl that replaces a full stop: the headline's end and the KR monogram.
`target-min`	`44px`	Minimum height and width of primary touch targets; secondary targets never go below 24px.

Stacking

One stacking map for the whole site.

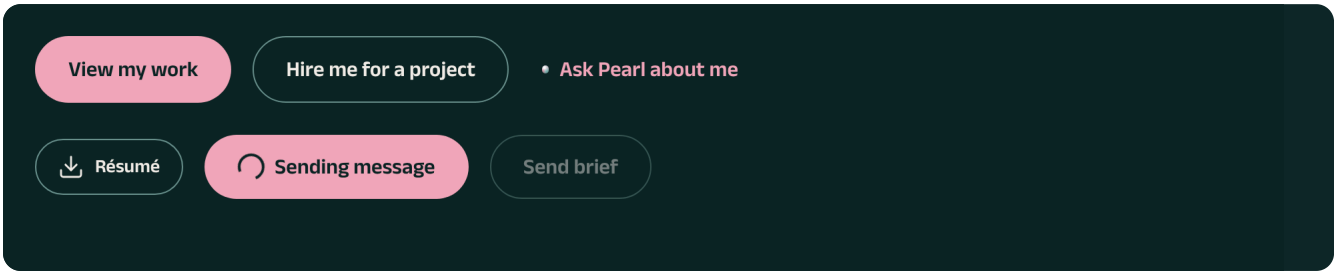
Token	Value	Use it for
`z-canvas`	`0`	The fixed WebGL canvas behind everything.
`z-content`	`1`	All page sections.
`z-nav`	`40`	The sticky navigation bar.
`z-launcher`	`50`	The pearl launcher.
`z-chat`	`60`	Pearl's panel and its mobile sheet.
`z-palette`	`70`	The command palette and its backdrop.
`z-toast`	`80`	Toasts, above everything a visitor can open.
`z-skip`	`100`	The skip-to-content link when focused.

Components

Each component as it renders in the Lagoon theme, followed by its contract. Plain-CSS references are in `components/bundle.css`; React prop types in `components/index.d.ts`.

Button

Primary, secondary, quiet, small, busy



Buttons start an action; a rose primary button appears at most once per view, for the thing that view exists to do.

Variants

- **primary**: `rose` fill with an `on-rose` label. "View my work" in the hero, "Send message" in Contact.
- **secondary**: no fill, 1.5 px `line-strong` border, `pearl` label. "Hire me for a project", "Résumé".
- **quiet**: `rose` text with no box, 44 px tall. Inline actions such as "Ask Pearl about me" and "Copy email".
- **size sm**: 40 px tall with a 14 px label, for the navigation's Résumé button and actions inside chat cards.

What the consumer provides

A verb-first label in sentence case (2 to 4 words), an optional 20 px leading icon from lucide-react, and either `href` (renders `<a>`) or `onClick` (renders `<button type="button">`). `busy` swaps the icon for a spinner and sets `aria-busy`; the label stays so the button doesn't change width.

States

Hover (`rose-hover` on primary, `shoal` on secondary), keyboard focus (`shadow-focus`), pressed (moves down 1 px), disabled (45% opacity with `aria-disabled`, so it stays focusable and the reason sits next to it), busy.

Tailwind recipe

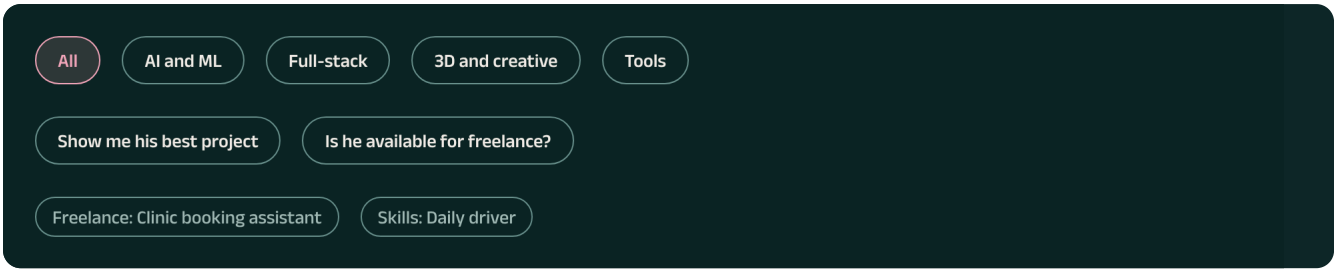
- Base: `inline-flex min-h-12 items-center justify-center gap-2 whitespace-nowrap rounded-pill px-6 text-base font-semibold transition-colors focus-visible:outline-none focus-visible:shadow-focus active:translate-y-px`
- Primary: `bg-rose text-on-rose hover:bg-rose-hover`
- Secondary: `border-[1.5px] border-line-strong text-pearl hover:bg-shoal`
- Quiet: `min-h-11 px-2 text-rose hover:text-rose-hover hover:underline underline-offset-4`

Do and don't

- Name the result, and keep that name through the flow: "Send message" leads to the toast "Message sent".
- Never append arrows ("View work →"), use all caps, or put two primary buttons side by side.
- Never build a button from a `div`; icon-only buttons need an `aria-label`.

Chip

Filter, starter, source



Chips are small pressable pills: project filters, Pearl's starter prompts, and the source links under Pearl's answers.

Kinds

- **filter**: toggles a project filter. Renders `<button aria-pressed>`. Pressed chips use `rose` text and border on `rose-soft`. Exactly one filter is pressed at a time ("All" by default).
- **starter**: a suggested question in Pearl. Pressing it sends the text as the visitor's message. Up to four at once, written as the visitor would ask ("Show me his best project").
- **source**: a link under a Pearl answer to the section it came from ("Freelance: Clinic booking assistant"). 28 px tall, `mist` text, 1 px `line-strong` border.

What the consumer provides

The label (sentence case, under 32 characters), plus `pressed` for filters, `onClick` for starters, or `href` for sources.

Tailwind recipe

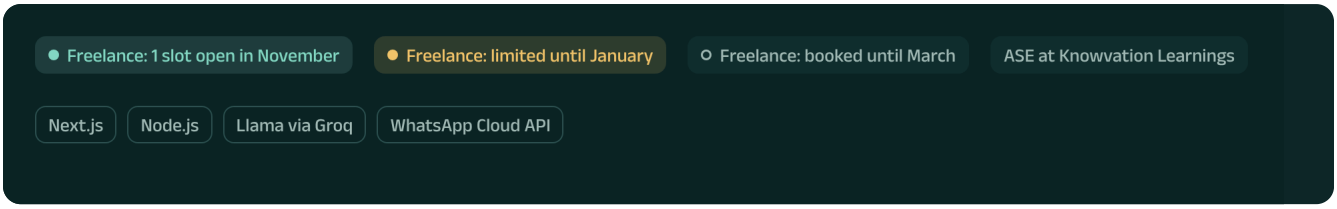
- Base: `inline-flex min-h-9 items-center gap-2 whitespace-nowrap rounded-pill border-[1.5px] border-line-strong px-4 text-sm font-medium text-pearl hover:bg-shoal focus-visible:outline-none focus-visible:shadow-focus`
- Pressed: `aria-pressed:border-rose aria-pressed:bg-rose-soft aria-pressed:text-rose`
- Source: `min-h-7 border px-3 text-mist hover:text-pearl`

Do and don't

- Chips are pills because they are pressable; read-only labels are Badges.
- Never use chips as a list of skills or tags; use Badge tags or plain text links.
- Keep a filter row to one line on desktop; let it scroll horizontally on phones rather than wrap into three rows.

Badge

Status tones and stack tags



Badges are read-only labels for live status and stack tags; they are rounded rectangles because nothing happens when you press them.

Tones

- **available**: `seaglass` on `seaglass-soft` with a filled dot. "Freelance: 1 slot open in November".
- **limited**: `amber` on `amber-soft` with a filled dot. "Freelance: limited until January".
- **booked**: `mist` on `reef` with a hollow dot. "Freelance: booked until March".
- **neutral**: `mist` on `reef`, no dot. Current employer in the hero: "ASE at Knowvation Learnings".
- **tag**: `mist` text with a 1 px `line` outline, no fill. Stack names on cards and timeline entries.

What the consumer provides

The words. Status tones always carry a word as well as colour, so they read the same in greyscale and to screen readers. The freelance status comes from one config value (`availability.freelance`), never hand-typed copy.

Tailwind recipe

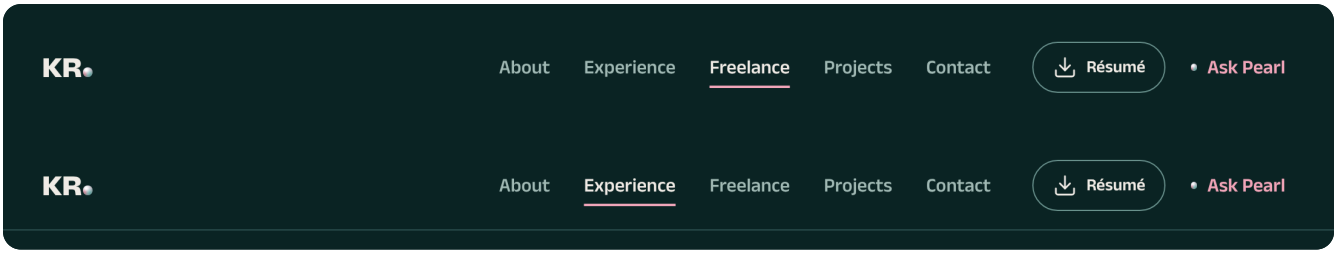
- Base: `inline-flex min-h-7 items-center gap-1.5 whitespace-nowrap rounded-sm px-2.5 text-sm font-medium`
- available: `bg-seaglass-soft text-seaglass` · limited: `bg-amber-soft text-amber` · booked and neutral: `bg-reef text-mist`
- tag: `text-mist ring-1 ring-inset ring-line`
- Dot: `size-2 rounded-full bg-current` (booked: `ring-[1.5px] ring-inset ring-current bg-transparent`)

Do and don't

- Never make a badge clickable; if it needs to act, it is a Chip.
- Never show more than five tags on a card; the case study lists the full stack.
- Status words are sentence case: "Available", never "AVAILABLE".

NavBar

Over the hero, and after scrolling



The navigation bar is a sticky strip with the KR monogram, five section links, the résumé, and a way into Pearl.

Anatomy

1. **Monogram**: "KR" in Anek Latin 780 at wdth 112, followed by the pearl dot. It links to the top. It is live type, not an image, so there is no logo file.
2. **Links**: About, Experience, Freelance, Projects, Contact, in `ui` at weight 560, `mist`; hover and the section in view turn `pearl`. The section in view gets `aria-current="location"` and a 2 px `rose` underline 10 px below the text.
3. **Actions**: Résumé (secondary, sm, download icon) and "Ask Pearl" (quiet, with the pearl dot).

Behaviour

- Transparent over the hero. After 80 px of scroll it takes `lagoon-veil`, a 12 px backdrop blur, and a `line` bottom border. The height never changes, so nothing shifts.
- Sticky at `top: env(safe-area-inset-top)`, `z-nav`. Sections set `scroll-margin-top: var(--nav-height)`.
- Below 860 px: monogram, Résumé, and a menu button. The menu opens a full-height sheet on `reef` (radius-lg top corners) with the links in `h3`, then Résumé, Ask Pearl, and socials. Focus is trapped in the sheet; Escape closes it and returns focus to the menu button.
- The first focusable element on the page is a "Skip to content" link above the bar (`z-skip`), visible only on focus.

What the consumer provides

`items` (label and hash), `activeId` from the section observer, `resumeHref`, and `onAskPearl`.

Tailwind recipe

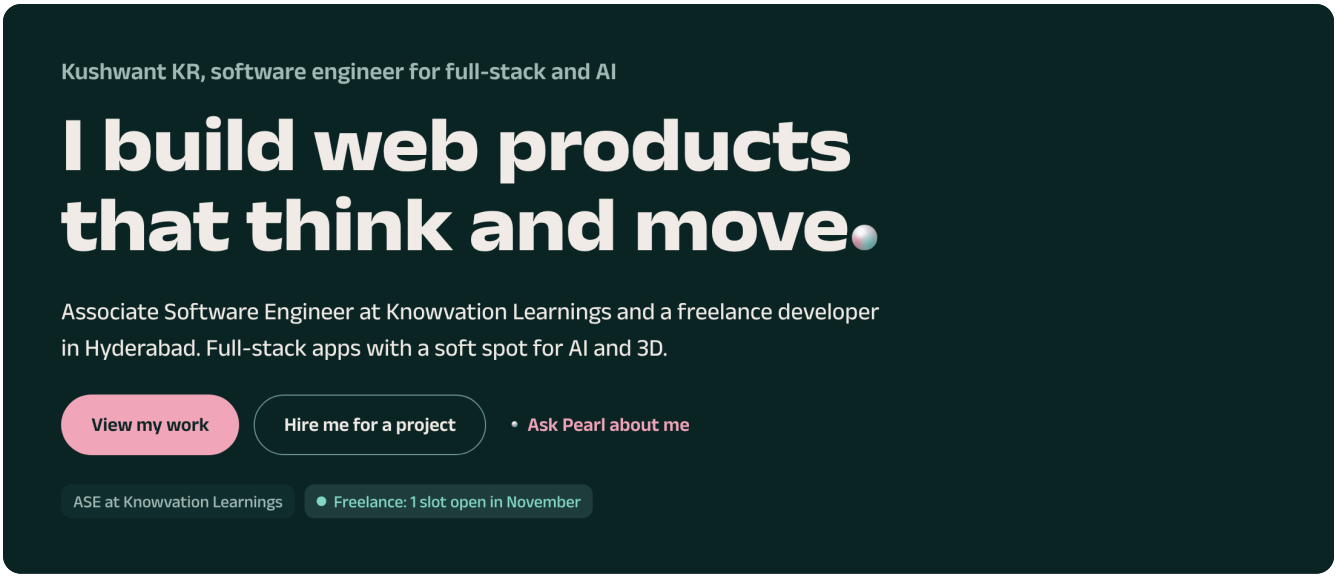
```
sticky top-[env(safe-area-inset-top)] z-nav flex h-18 max-[860px]:h-16 items-center gap-8 px-8 max-[860px]:px-4 data-[scrolled=true]:bg-lagoon-veil data-[scrolled=true]:backdrop-blur-md data-[scrolled=true]:border-b data-[scrolled=true]:border-line
```

Do and don't

- Keep it to five links; Skills and Education are reached by scrolling.
- Never hide the Résumé button on any screen size.

Hero

Text block with the pearl full stop



The hero's text block states who Kushwant is and what he does, and ends its headline with the pearl instead of a full stop.

Anatomy

1. **Name line**: "Kushwant KR, software engineer for full-stack and AI", in `lead` size at weight 600, `mist`.
2. **Headline**: `display` (wdth 114), `pearl`, balanced over two or three lines. The final full stop is a `.kr-pearl` span (0.32em), marked `aria-hidden`; the sentence keeps a real full stop for screen readers in a visually hidden span.
3. **Sub-headline**: `lead`, `pearl`, max 60ch.
4. **Actions**: primary "View my work", secondary "Hire me for a project", quiet "Ask Pearl about me".
5. **Status**: a neutral badge for the day job and a status badge for freelance availability.

Name line and headline sit inside one `<h1>`, so search engines and screen readers get both the name and the claim.

Layout with 3D

On desktop the text block spans columns 1 to 7 of the 12-column grid and the 3D pearl sits just past the end of the headline, around columns 8 to 12, so the eye travels from the words into the object. On phones the pearl sits above the text, cropped at the top-right, and never behind it. The block is in the HTML from the first byte: the headline is the LCP element, and the 3D scene never hides or delays it.

The opening (one orchestrated moment)

When the canvas is ready, the WebGL pearl appears exactly over the CSS full stop, at its size, then grows and travels to its hero position over 1,200 ms while the particles condense into their orbit. The CSS full stop stays in place: the big pearl grew out of it. This plays once per session and never under reduced motion. Nothing else on the page animates in.

What the consumer provides

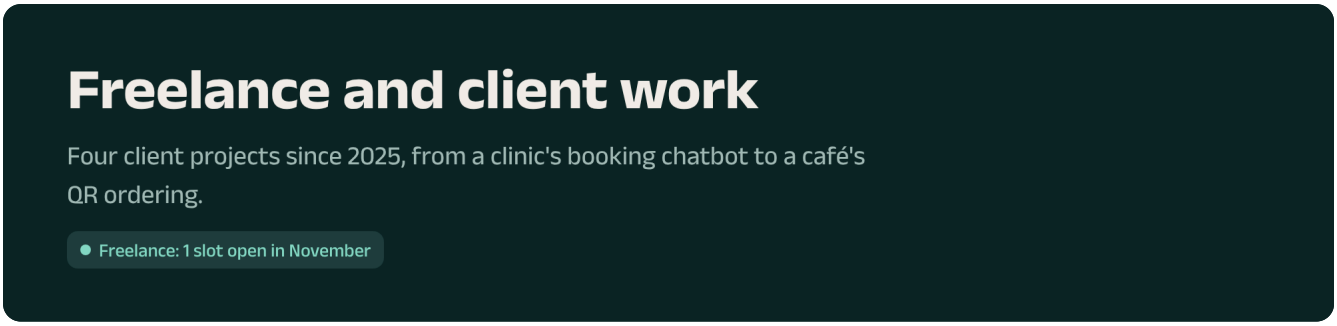
`name`, `role`, `headline` without its final full stop, `subheadline`, both CTAs, `onAskPearl`, `employment`, `availability`, and a `pearlAnchorRef` on the full-stop span so the 3D layer can read its screen position.

Do and don't

- One `display` headline per site. Case-study pages start at `h1`.
- Never put a second accent colour, gradient text, or a typing effect in the headline.
- Never size the hero to more than `hero-min-height`; the Experience section should peek on tall screens.

SectionHeader

Title, summary, optional aside



Every home-page section opens with its title and a one-line summary of what the visitor will find, and nothing above the title.

Anatomy

- **Title**: `<h2>` in `h2` (wdth 108), `pearl`, balanced. Written as a plain description: "Experience", "Freelance and client work", "Projects", "Skills", "Education and achievements", "Let's build something".
- **Summary**: `lead`, `mist`, max 60ch. One sentence that carries the section's point, with a number where there is one: "Four client projects since 2025, from clinics to cafés."
- **Aside** (optional): one element aligned to the summary's baseline on desktop, such as the availability badge in Freelance.

The section element carries the `id` used by the navigation (`#freelance`) and `scroll-margin-top: var(--nav-height)`. The gap to the content below is `space-7`.

Tailwind recipe

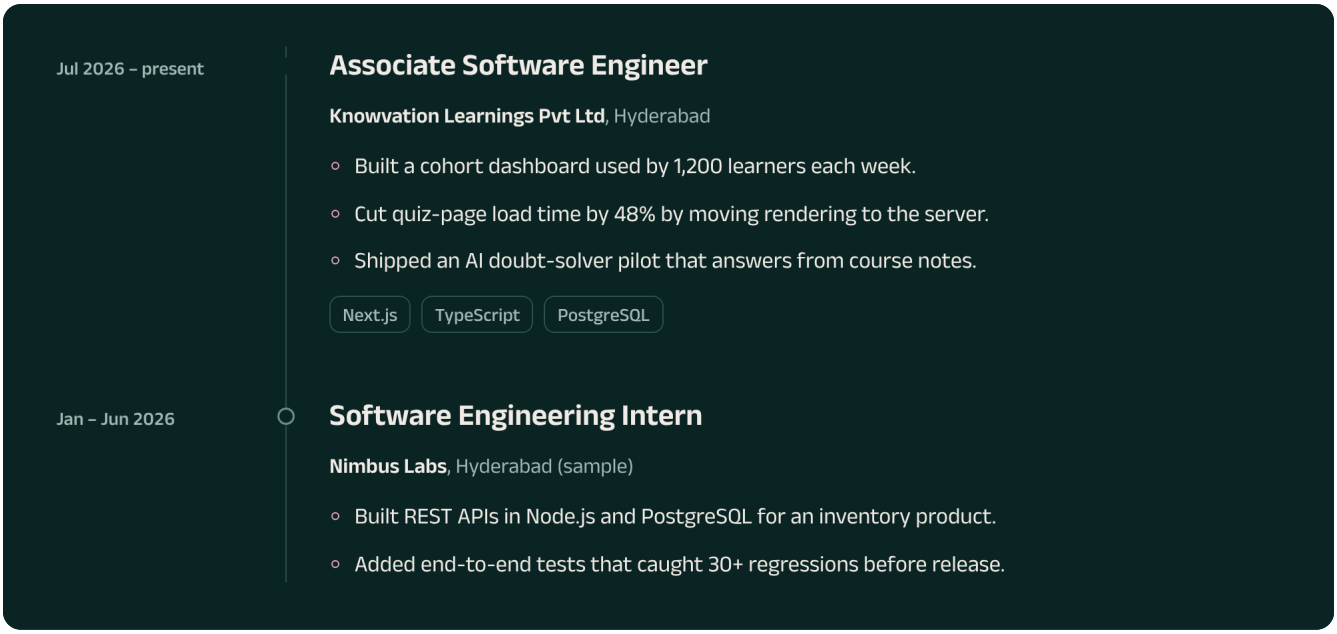
`grid max-w-[68ch] gap-3` with the title `text-h2 font-bold tracking-tight [font-variation-settings:'wdth'_108] text-balance` and the summary `text-lead text-mist max-w-[60ch]`.

Do and don't

- Never put an eyebrow label, a number ("01"), or an all-caps kicker above the title.
- Never colour or italicise one word of a title.
- Never centre a section header; everything aligns to the left edge of the grid.

TimelineItem

Current role with the pearl mark, then a past role



A timeline item is one role in the Experience section, hung on a growth line like a layer of nacre, newest first.

Anatomy

- **When** (left column, 148 px): "Jul 2026 – present" in `small` at weight 560, `mist`, tabular figures, inside `<time dateTime>` elements.
- **Mark**: a 13 px dot on the 1 px `line` growth line. The current role's mark is the CSS pearl; past roles are hollow `line-strong` rings.
- **Role**: `h4` (`<h3>` in the outline), `pearl`.
- **Organisation**: organisation name in `pearl` weight 600, then location in `mist`: "Knowvation Learnings Pvt Ltd, Hyderabad".
- **Points**: 3 to 5 outcome bullets in `body`, `rose` markers, each leading with a verb and a number where one exists.
- **Stack**: up to five tag badges.

Entries are separated by `space-7` of growth line. Below 640 px the date moves above the role, on the same line.

What the consumer provides

`role`, `org`, `location`, ISO `start` and optional `end` (absent means present), `points`, `stack`, and `current` for the pearl mark. Format dates as "Jul 2026", with an en dash between start and end.

Tailwind recipe

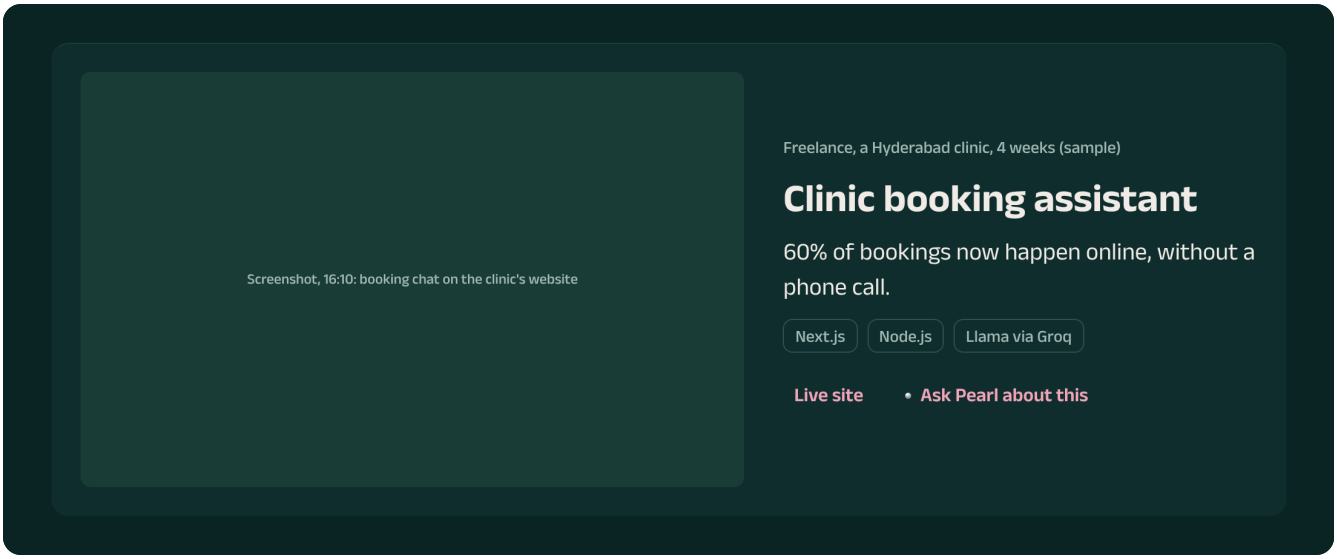
Row: `grid grid-cols-[148px_minmax(0,1fr)] gap-x-6 max-sm:grid-cols-1`. Body: `relative grid gap-3 border-l border-line pb-12 pl-8 last:pb-0`. Mark: `absolute -left-[7px] top-2 size-[13px] rounded-full ring-[1.5px] ring-inset ring-line-strong bg-lagoon`.

Do and don't

- Only true sequences get a timeline; this is one, so the dates are the markers. No "01, 02" numbering.
- Never list duties ("Responsible for..."); write outcomes ("Cut quiz-page load time by 48%").
- Internships and the current role share this component; freelance work never appears here, it has its own section.

CaseStudyCard

Freelance case study row



A case-study card is a wide row that sells one project by its result, used for both featured projects and freelance work.

Anatomy

- **Media** (7 of 12 columns): a 16:10 screenshot or muted looping video on `shoal`, `radius-sm`, inside the card's `reef` mat. Video plays only on desktop and only while in view; phones get the poster.
- **Context**: `small`, `mist`, commas between parts: "Freelance, a Hyderabad clinic, 4 weeks" or "Personal project, 2025".
- **Title**: `h3` (`<h3>`), linking to the case study. Its link stretches over the whole card with a pseudo-element, so the card is one target.
- **Outcome**: one sentence in `lead`, `pearl`, leading with the result and its number: "60% of bookings now happen online, without a phone call."
- **Stack**: up to five tag badges.
- **Links**: quiet buttons such as "Live site", "Code", "Ask Pearl about this", raised above the stretched link so they stay separately clickable.

Rows alternate media side (`flip`) down a list on desktop; below 768 px the media always sits on top.

States

Hover lifts the mat from `reef` to `shoal`. Keyboard focus on the title link draws `shadow-focus` around the whole card. No scale, tilt, or shadow growth on hover.

What the consumer provides

`kind`, `context`, `title`, `href`, `outcome`, `media` (with alt text that says what the screenshot shows), `stack`, optional `links`, `flip`, and `onAskPearl`.

Tailwind recipe

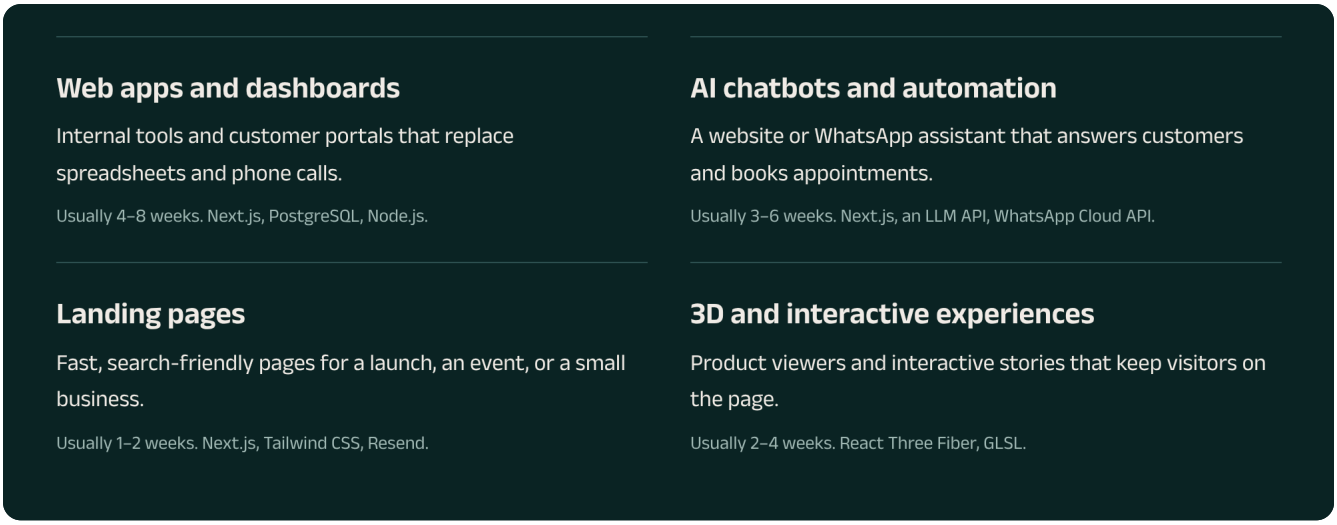
```
relative grid grid-cols-[minmax(0,7fr)_minmax(0,5fr)] items-center gap-8 rounded-md bg-reef p-6 shadow-lustre transition-colors hover:bg-shoal focus-within:shadow-focus max-md:grid-cols-1
```

Do and don't

- Freelance and personal projects use the same card, but the context line always says which it is.
- Never invent a metric. If there is no number yet, lead with the concrete result ("Doctors confirm bookings from WhatsApp").
- Never use device-frame mockups or stock images; real screenshots only.

ServiceList

Four services as hairline rows



The service list tells a freelance client what Kushwant builds, how long it usually takes, and with what, as hairline rows instead of a grid of boxes.

Anatomy

Each service is a row topped by a 1 px `line` rule:

- **Name**: `h4`, `pearl`: "AI chatbots and automation".
- **Description**: one sentence in `body`, max 46ch, from the client's side of the table: "A website or WhatsApp assistant that answers customers and books appointments."
- **Meta**: `small`, `mist`, as a sentence: "Usually 3–6 weeks. Next.js, Node.js, an LLM API." When `startingFrom` is enabled in config, it follows: "From ₹40,000."

Two columns on desktop, one below 768 px. Three or four services; never more.

What the consumer provides

`services` with `name`, `description`, `timeline`, `stack`, and optional `startingFrom` (hidden unless `config.showPrices` is true).

Tailwind recipe

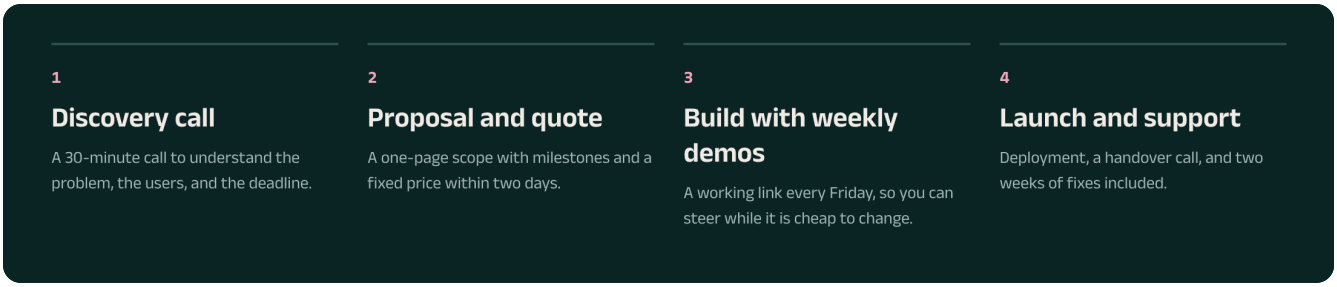
List: `grid grid-cols-2 gap-x-8 max-md:grid-cols-1`. Row: `grid content-start gap-2 border-t border-line py-6`.

Do and don't

- Rupee amounts use Indian digit grouping: ₹1,50,000.
- Never add icons or illustrations per service; the words carry it.
- Never show a price that is not in config; Pearl quotes only published prices.

ProcessSteps

Four ordered steps



Process steps show a client the four stages of working with Kushwant; it is numbered because the order is the information.

Anatomy

An ordered list (`<ol>`) of four steps, each topped by a 2 px `line` rule:

- **Number**: generated by a CSS counter in `small` weight 640, `rose`, tabular figures.
- **Name**: `h4` : "Discovery call", "Proposal and quote", "Build with weekly demos", "Launch and support".
- **Description**: one sentence in `small` , `mist` : "A 30-minute call to understand the problem, the users, and the deadline."

Four columns on desktop, two below 860 px, one below 480 px.

What the consumer provides

`steps` with `name` and `description` , in order.

Tailwind recipe

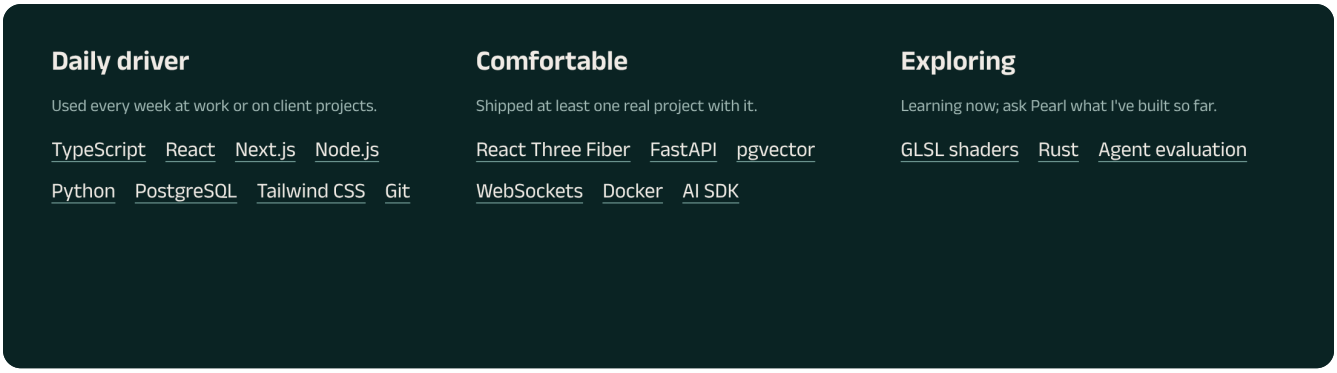
List: `grid grid-cols-4 gap-6 max-[860px]:grid-cols-2 max-[480px]:grid-cols-1 [counter-reset:step]` . Step: `grid content-start gap-2 border-t-2 border-line pt-4 [counter-increment:step] before:content-[counter(step)] before:text-sm before:font-semibold before:text-rose` .

Do and don't

- Use numbers only here and in other true sequences. Services, projects, and skills are never numbered.
- Keep each description to one sentence; details belong in the proposal, not the page.

SkillTiers

Daily driver, comfortable, exploring



Skill tiers sort technologies into three honest columns, each skill linking to the work that proves it, instead of percentage bars.

Anatomy

- **Tier name**: `h4` : "Daily driver", "Comfortable", "Exploring".
- **Tier note**: one line in `small` , `mist` , defining the tier: "Used every week at work or on client projects." / "Shipped at least one real project with it." / "Learning now; ask Pearl what I've built so far."
- **Skills**: an inline list of text links in `body` , `pearl` , underlined in `line-strong` ; hover turns them `rose` . Each link goes to the strongest evidence (a case study, a timeline entry, a repo).

Three columns on desktop, stacked below 768 px.

What the consumer provides

`tiers` , each with `name` , `note` , and `skills` as `{ name, evidenceHref }` . Every Daily-driver skill must have an evidence link; the content check fails the build otherwise. Pearl reads the same data for "Has he used X?" answers.

Tailwind recipe

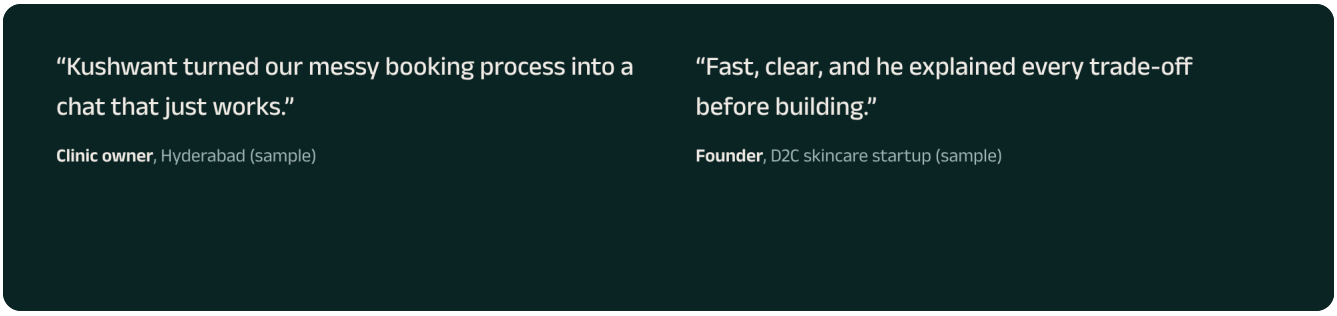
Grid: `grid grid-cols-3 gap-8 max-md:grid-cols-1` . Skill link: `text-pearl underline decoration-line-strong decoration-1 underline-offset-4 hover:text-rose hover:decoration-rose` .

Do and don't

- Never show percentages, stars, or progress bars; they claim a precision nobody has.
- Never pad the lists with every library ever touched; eight to twelve skills per tier at most.
- Never use logos for skills; names are clearer and faster to scan.

Testimonial

Two quotes at the end of Freelance



A testimonial is a client's or colleague's own words about working with Kushwant, set as a plain quotation with its source.

Anatomy

`<figure>` with a `<blockquote>` in `lead` at weight 500, `pearl`, max 56ch, and a `<figcaption>` in `small`, `mist`: the person's name in `pearl` weight 600 (when they allow it), then role and organisation: "Dr. A. Rao, clinic owner, Hyderabad".

Place testimonials inside the case study they belong to, or two side by side at the end of the Freelance section. They sit directly on `lagoon`, with no card.

What the consumer provides

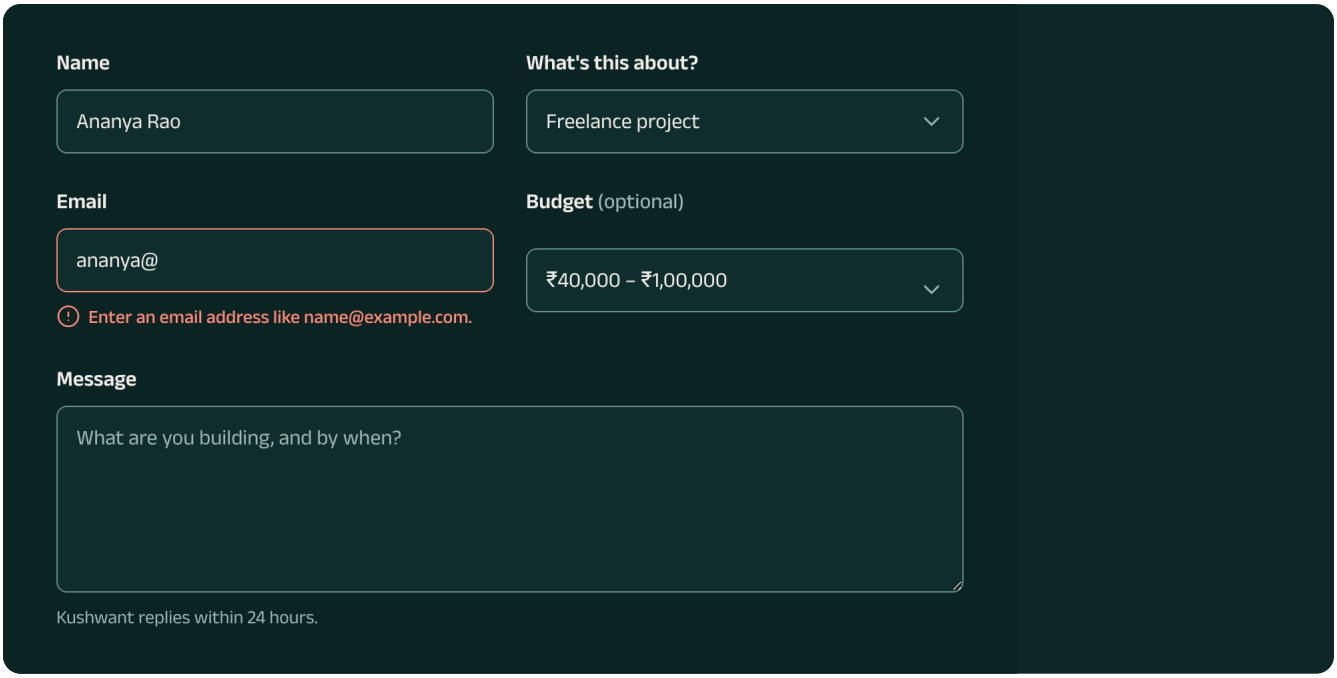
`quote` (the person's exact words, lightly trimmed with an ellipsis where shortened), `role`, and optionally `name` and `org`. A testimonial with `placeholder: true` never ships to production.

Do and don't

- Use only quotes the person approved in writing; keep the approval with the case-study file.
- Never add star ratings, avatars generated from initials, or giant decorative quote marks.
- Never rewrite a quote to sound better; shorten it instead.

FormField

Default, select, invalid, textarea



A form field keeps a label, its control, an optional hint, and an error together, so the contact form reads and validates as one unit.

Anatomy

- **Label**: `ui` (16 px, 600), `pearl`, always visible above the control; optional fields add " (optional)" in `mist`. Never a placeholder-only label.
- **Control**: 48 px minimum height, 16 px text (no zoom on iOS), `reef` fill, 1.5 px `line-strong` border, `radius-sm`. Textareas start at 140 px and grow. Selects draw a 20 px chevron in `mist`.
- **Hint**: `small`, `mist`, under the control: "We reply within 24 hours."
- **Error**: `small` weight 520, `coral`, with an alert icon, replacing the hint: "Enter an email address like name@example.com."

States

Hover darkens the border to `mist`; focus turns it `rose` and adds `shadow-focus`; invalid turns it `coral` and sets `aria-invalid` with `aria-describedby` pointing at the error. Errors appear after the field loses focus or on submit, never while the visitor is still typing.

The contact form

Fields in order: Name, Email, What's this about? (Job opportunity, Freelance project, Collaboration, Just saying hi), Message. Choosing Freelance project reveals Project type, Budget (₹ ranges plus "Not sure yet"), and Timeline. The same zod schema validates in the browser and in the server action, and the form posts without JavaScript.

Tailwind recipe

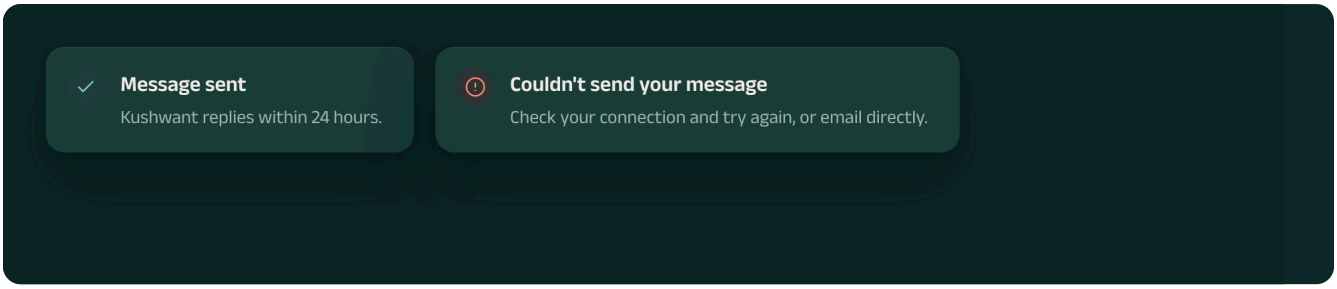
Control: `min-h-12 w-full rounded-sm border-[1.5px] border-line-strong bg-reef px-3.5 py-2.5 text-base text-pearl placeholder:text-mist hover:border-mist focus-visible:border-rose focus-visible:outline-none focus-visible:shadow-focus aria-invalid:border-coral`.

Do and don't

- Errors say what to do, not what went wrong in system terms: never "Invalid input" or "Error 422".
- Never disable the submit button to signal errors; let the visitor submit and show every error at once, with focus moved to the first.

Toast

Success and error



A toast confirms that something the visitor did has happened, or says plainly that it failed and what to do next.

Anatomy

A `shoal` surface with `radius-md`, `shadow-float`, and `shadow-lustre`, max 420 px wide: a 28 px round icon (check on `seaglass-soft` for success, alert on `coral-soft` for errors), a title in `ui`, and an optional line in `small`, `mist`.

Placement and timing

Bottom-left on desktop (the launcher owns bottom-right), bottom-centre above the safe area on phones, at `z-toast`. Success toasts leave after 5 s; error toasts stay until dismissed. Announced through an `aria-live="polite"` region (errors use `role="alert"`). Enter by fading and rising 8 px over 240 ms; reduced motion fades only.

Copy

The title repeats the action's own words in the past tense:

- "Message sent" / "Kushwant replies within 24 hours."
- "Email copied" / "kushwant@... is on your clipboard."
- "Couldn't send your message" / "Check your connection and try again, or email directly."

Do and don't

- Never stack more than two toasts.
- Never use a toast for something the visitor must read later; put it on the page.

PearlLauncher

Idle with label, thinking, unread



The pearl launcher is the floating button that opens Pearl; it is the same pearl as the hero's 3D object and the headline's full stop.

Placement

Fixed bottom-right, 56 px (`launcher-size`), 48 px on phones. Offsets: `right: max(24px, env(safe-area-inset-right) + 16px)` and `bottom: max(24px, env(safe-area-inset-bottom) + 16px)` . Stacks at `z-launcher` . It appears once the hero has scrolled half out of view; on the hero itself, the hero's "Ask Pearl about me" link does the job.

Two renderings, one button

- **With 3D on**, the WebGL pearl flies from the hero into this corner (the handoff, P1) and is drawn underneath a transparent button; pass `webglPearl` so the CSS pearl hides.
- **With 3D off, low tier, or before the canvas loads**, the button paints the CSS pearl (`scene-pearl` body, `scene-sheen-rose` and `scene-sheen-aqua` edges, a white highlight). The two must sit at the same size and position, so switching tiers never moves the target.

States

- **idle**: hover scales to 1.06.
- **thinking**: Pearl is answering while the panel is closed; a slow hue shimmer (1.6 s). Reduced motion swaps it for a static `seaglass` ring.
- **open**: scales to 0.92 while the panel is open (desktop). On phones the sheet covers it.
- **unread**: a 12 px `rose` dot at the top-right when Pearl replied while closed.

Accessibility

`<button aria-label="Ask Pearl, Kushwant's AI guide" aria-expanded aria-controls="pearl-panel">` . A visible "Ask Pearl" label appears beside it on hover and keyboard focus after 400 ms. Escape closes the panel and returns focus here.

Tailwind recipe

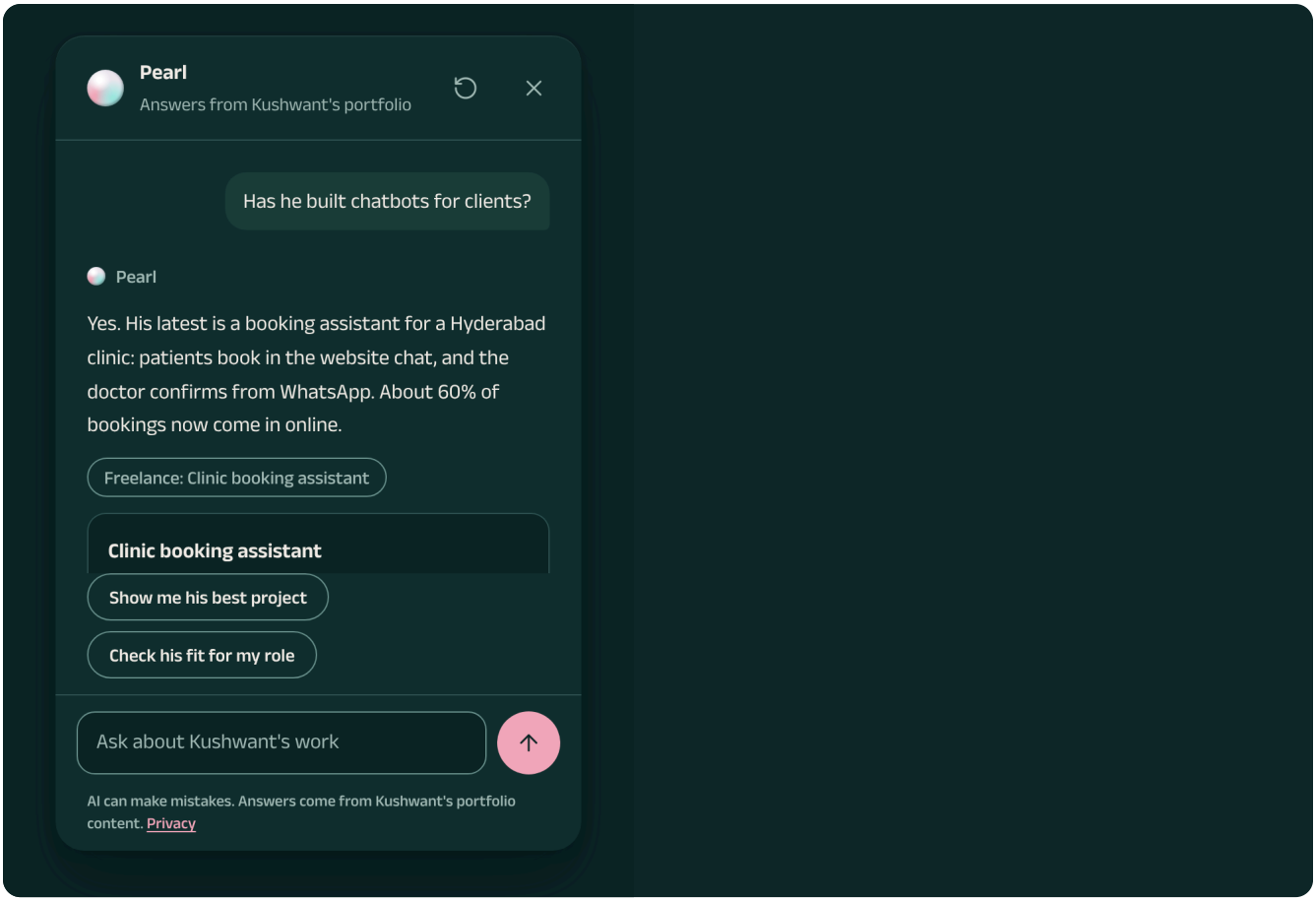
`fixed z-launcher size-14 max-sm:size-12 rounded-full shadow-float focus-visible:outline-none focus-visible:shadow-focus` plus the `.kr-pearl` background utility from the Implementation section.

Do and don't

- Never add an icon or text on the pearl itself; the pearl is the icon.
- Never bounce, wiggle, or pulse it to get attention.

ChatPanel

Answer with sources and a card, then the confirm step



The chat panel is where visitors talk to Pearl: a 400 px side panel on desktop and a 90svh bottom sheet on phones, with the page still usable behind it on desktop.

Anatomy

1. **Header**: a 28 px pearl, "Pearl" in `ui` 640, a subline in `small` `mist` ("Answers from Kushwant's portfolio"), then New chat and Close icon buttons (40 px, labelled).
2. **Log** (`role="log"` , `aria-live="polite"`): messages separated by `space-5` . - **Visitor**: a `shoal` bubble on the right, `radius` 16 16 4 16, max 85%. - **Pearl**: no bubble. A meta line (14 px pearl and the word "Pearl" in `mist`), then plain paragraphs in `pearl` , then parts.
3. **Parts** inside Pearl's answers: source chips (links to the section the answer came from), a project card, a résumé card, a job-description fit card, and the confirm card that shows exactly what will be sent to Kushwant with Send and Edit buttons.
4. **Starters**: up to four chips above the composer. At the start they come from the section in view; after an answer, they become two follow-ups.
5. **Composer**: an auto-growing textarea (48 to 140 px) on `lagoon` with `radius-md` , and a 48 px round `rose` send button. Enter sends, Shift+Enter adds a line.
6. **Footer**: `caption` , `mist` : "AI can make mistakes. Answers come from Kushwant's portfolio content." with a link to /privacy.

States

- **streaming**: three 8 px pearls bob in the meta line until the first token arrives; text then streams in. The hero pearl shimmers at the same time when 3D is on.
- **error**: an inline coral line under the last message: "Pearl couldn't answer that. Try again." with a Retry quiet button.
- **rate-limited**: an amber notice: "You've asked a lot of questions in a few minutes. Pearl can answer again at 8:40 PM." plus Copy email.
- **offline**: when the model is down or the daily budget is spent, the subline changes to "Quick answers mode" and Pearl answers from the local FAQ, with contact options in every reply.

Behaviour

Opening moves focus to the composer; Escape or Close returns it to whatever opened the panel. On phones the sheet traps focus and the page behind it is inert. Conversations last for the browser session; New chat clears them. Opening spring: stiffness 380, damping 34, from 24 px below with opacity 0; reduced motion uses a 120 ms fade.

What the consumer provides

`id`, `messages` (AI SDK UI messages mapped to `PearlPart`s), `status`, `starters`, `onSend`, `onClose`, `onNewChat`.

Do and don't

- Never show model names, token counts, or "As an AI language model" copy.
- Never send anything to Kushwant without the confirm card and a visitor click.
- Never put Pearl's answers in coloured bubbles; plain text keeps long answers readable.

CodeBlock

Excerpt from the particle morph

src/scene/Particles.tsx Copy

```
// Ease toward the formation the scroll position asks for
useFrame((state, delta) => {
  const u = material.uniforms
  easing.damp(u.uProgress, 'value', scroll.progress, 0.25, delta)
  u.uPointer.value.lerp(state.pointer, 0.08)
  u.uTime.value += delta
})
```

The morph runs in `useFrame`, so scrolling never triggers a React render.

Code blocks show short, real excerpts in case studies and in Pearl's developer answers, in Martian Mono on a `reef` panel.

Anatomy

- **Header**: the file path in `small`, `mist` ("src/scene/Particles.tsx"), and a quiet sm "Copy" button that becomes "Copied" for 2 s.
- **Body**: `code` style (Martian Mono 15 px, wdth 87.5, line height 1.6), `pearl`, padded `space-4` by `space-5`, scrolling horizontally inside its own container.
- **Highlighting** maps to tokens: keywords `rose`, strings `seaglass`, numbers `amber`, comments `mist`. Everything else stays `pearl`. All four pass 5.8:1 on `reef` in both themes.
- **Inline code** in running text: `code-inline` on `reef`, `radius-xs`.

What the consumer provides

`code`, `language`, and an optional `filename`. Highlight at build time (Shiki with a theme generated from these tokens), never in the browser.

Do and don't

- Keep excerpts under 25 lines; link to the repo for the rest.
- Never use the mono face for labels, dates, or metadata.

Layout

Every section is left-aligned on one 12-column grid, sits directly on `lagoon`, and leaves the 3D layer room to breathe on the right.

Grid and breakpoints

Breakpoint	Width	Columns	Column gap	Side gutter	Section padding
Phone	below 640 px	4	16 px	`max(16px, env(safe-area-inset-left))`	`space-8` (64 px)
Tablet (sm, md)	640–1023 px	8	24 px	32 px	`space-9` (96 px)
Desktop (lg, xl)	1024 px and up	12	24 px	48 px	`space-10` (128 px)

- Content never exceeds `container-max` (1200 px); the canvas and section backgrounds run full width.
- Running text stops at `measure` (68ch). Sections set `scroll-margin-top: var(--nav-height)`.
- Lay out siblings with `gap`. Never stack margins between sections.

Home page, section by section

Each wireframe shows desktop, then the phone change. Brackets are buttons or badges; the dot (•) is the pearl.

Hero

```
KR•                     About  Experience  Freelance  Projects  Contact   [Résumé]  •Ask Pearl

Kushwant KR, software engineer for full-stack and AI
I build web products                              .  .  .  .  .
that think and move•  – – – – – – – – – ▶     ·    ( PEARL )    ·     orbit of particles
                                                  '  .  '  .  '
Associate Software Engineer at Knowvation Learnings and a freelance
developer in Hyderabad. Full-stack apps with a soft spot for AI and 3D.
(View my work)  (Hire me for a project)  •Ask Pearl about me
[ASE at Knowvation Learnings]  [• Freelance: 1 slot open in November]

text: columns 1–7                                 3D: columns 8–12, centred on the headline's end
```

Phone: the pearl sits above the text, cropped at the top-right at about 60% of the screen width; the text never overlaps it. Height is `hero-min-height`.

About

```
About
A 98-word bio in body, max 68ch.                [portrait, duotone, 4:5, P1]
Based in Hyderabad   B.Tech CSE, 2026   ASE at Knowvation   4 freelance projects
What I'm into: LLM apps and retrieval, real-time 3D on the web, developer tooling
Now: building Pearl, learning GLSL shaders
```

The quick facts are a definition list laid out in a row that wraps; separate them with space, not with dots.

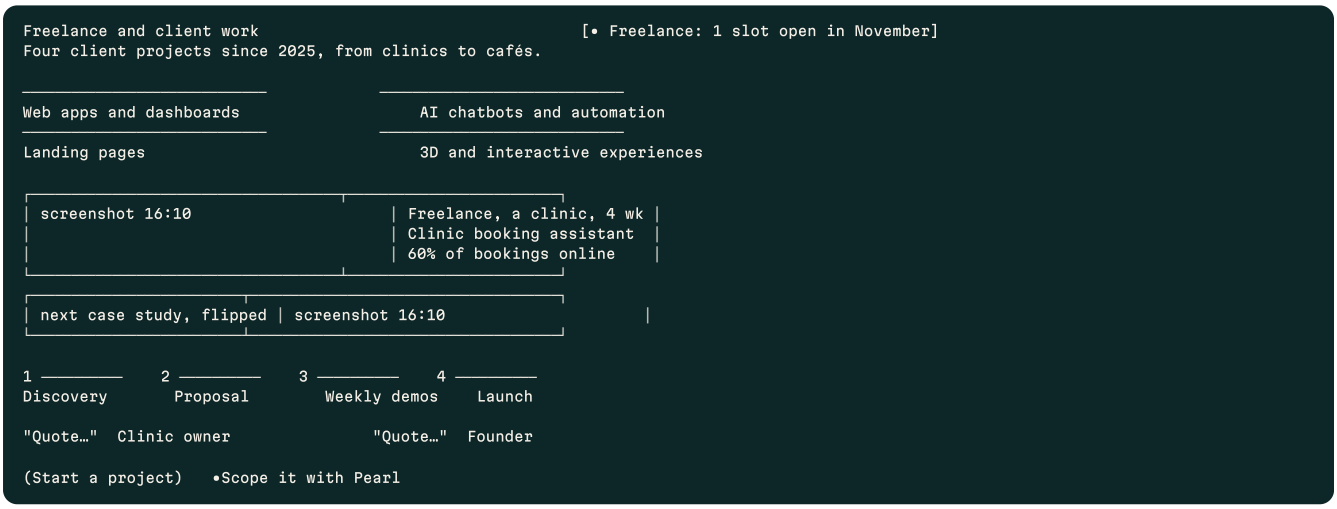
Experience

```
Experience
Summary line in lead.

Jul 2026 – present   •  Associate Software Engineer
                     |  Knowvation Learnings Pvt Ltd, Hyderabad
                     |  • Built a cohort dashboard used by 1,200 learners
                     |  [Next.js] [TypeScript] [PostgreSQL]
                     |
Jan – Jun 2026       ○  Software Engineering Intern
```

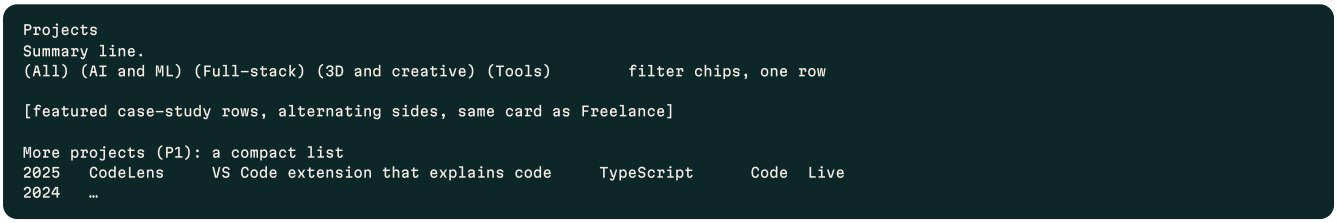
Phone: the date moves above each role.

Freelance



Phone: services stack, case-study media goes on top, the process becomes one column.

Projects

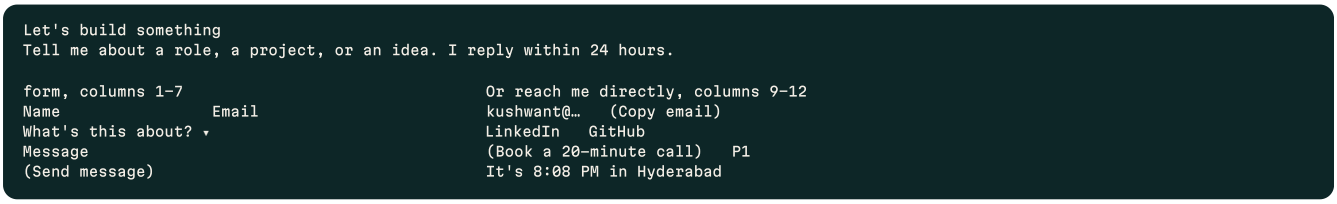


Phone: the filter row scrolls sideways; the archive list stacks into two lines per project.

Skills, Education and achievements



Contact



Footer

Mini navigation, socials, "How this site is built", "Last updated 22 Oct 2026", the Motion and 3D toggles, and "© 2026 Kushwant KR". One row on desktop, stacked on phones, `small` in `mist`.

Pearl's panel

- **Desktop**: fixed to the right at `chat-panel-width`, `top: calc(var(--nav-height) + 16px)`, `bottom` and `right` 24 px plus safe-area insets, `z-chat`. The page behind stays scrollable and clickable.
- **Phone**: a bottom sheet of 90svh with `radius-lg` top corners, a 36 by 4 px `line-strong` drag handle, a `lagoon-veil` backdrop, and the page made inert.

Case-study pages

A single 8-column reading column (columns 3–10 on desktop): breadcrumb ("Work / Clinic booking assistant"), `h1`, the outcome in `lead`, a facts row (role, timeline, team, stack), a full-width 16:10 hero image (columns 1–12), then sections in `h2`: The problem, What I built, Key decisions, Results, What I'd do differently. Code blocks and diagrams may run columns 2–11. Previous and next links close the page.

Stacking

`z-canvas` 0 for the fixed canvas, `z-content` 1, `z-nav` 40, `z-launcher` 50, `z-chat` 60, `z-palette` 70, `z-toast` 80, `z-skip` 100. Nothing uses a z-index outside this list.

Motion and 3D

The scene is one pearl in a field of particles that rearranges itself as the visitor scrolls; every value below is a starting point to tune by eye, not a law.

Motion tokens

Token	Value	Use
`--duration-press`	100ms	Press feedback (1 px down)
`--duration-quick`	160ms	Hover colour changes, launcher scale
`--duration-base`	240ms	Menus, toasts, chips, the nav background
`--duration-panel`	320ms	Chat panel and sheet when springs are unavailable
`--duration-scene`	1200ms	The hero opening move
`--ease-out`	cubic-bezier(0.22, 1, 0.36, 1)	Entrances and hovers
`--ease-in-out`	cubic-bezier(0.65, 0, 0.35, 1)	Morphs and the opening move
spring `panel`	stiffness 380, damping 34	Chat panel and mobile sheet (Motion)
damp `scroll`	smoothTime 0.25 s	Formation progress (`maath` `easing.damp`)
lerp `pointer`	0.08 per frame	Pointer influence on particles and the pearl's highlight

Motion tokens live in CSS and in `src/lib/motion.ts`; the design system has no motion token family, so this table is the source.

What moves, and when

- **Unprompted**: the hero opening (once per session) and the particle formations following scroll. Nothing else.
- **In answer to a visitor**: hovers, presses, the chat panel opening, Pearl's typing dots, the launcher's shimmer while Pearl answers, toasts.
- **Never**: section fade-ups on scroll, staggered card entrances, parallax text, cursor trails, custom cursors, marquee text.

The opening sequence

Moment	What happens
First paint	The name, the headline with its CSS pearl full stop, and the buttons are already visible. Nothing waits for JavaScript.
Canvas ready (typically 0.3–0.9 s)	The canvas fades in over 400 ms; the WebGL pearl appears exactly over the full stop, at its size.
Next 1,200 ms	The pearl grows to hero size and travels to its place on `--ease-in-out`. The particles condense from a wide scatter into the orbit ring over 1,400 ms.
After	Idle: the pearl turns once every 40 s, the particles drift slowly, the highlight follows the pointer.

Skip the sequence when `prefers-reduced-motion` is set or `sessionStorage` holds `kr-opened`; the scene then starts in its idle state.

Formations

Every particle stores a position in each of the five formations; the vertex shader blends between them with weights driven by scroll. Each section's first 35% of scroll blends from the previous formation with a smoothstep.

Formation	Sections	Shape	Parameters	Opacity
Orbit	Hero	Tilted ring around the pearl	Radius 2.6, tube spread σ 0.35, tilt 20°; 15% of particles in a halo at radius 4–6	100%
Layers	About, Experience	Concentric shells, like nacre in cross-section	Radii 1.3, 1.7, 2.1, 2.5, 2.9; jitter 0.03	80%
Clusters	Freelance	One cluster per service	Gaussian σ 0.45; centres on an arc from x −3 to 3	80%
Field	Projects, Skills, Education	Sparse dust behind the content	Uniform box 14 × 8 × 6	35%
Wave	Contact, footer	A calm swell	Sheet 14 × 6 at y −1.6; amplitude 0.15, speed 0.4	60%

The pearl

Property	High and Medium	Low
Geometry	Sphere, radius 1, 128 segments (64 on Medium)	48 segments
Material	`MeshPhysicalMaterial`	Matcap baked from the High pearl (128 px), or `MeshStandardMaterial` until it exists
Colour	`scene-pearl`	`scene-pearl`
Roughness / metalness	0.18 / 0	0.22 / 0.05
Clearcoat	1, roughness 0.08	none
Iridescence	1, IOR 1.3, thickness 200–600 nm	none
Sheen	0.4, colour `scene-sheen-rose`, roughness 0.5	none
Environment intensity	1.2	1.0

Placement: desktop x +1.9, y +0.2, scale 1; phones x +1.1, y +1.6, scale 0.75. During the handoff (P1) its target is the launcher's screen position unprojected onto the z = 0 plane, scaled to 56 px.

Light and colour management

- No HDR download. A procedural environment (`Environment`, resolution 256) holds three light panels: a key panel top-left (intensity 2.2, 5 × 2, white), a rim ring behind-right (intensity 1.4, `scene-sheen-aqua`), and a warm fill below-right (intensity 0.6, `scene-sheen-rose`).
- AgX tone mapping at exposure 1.0, sRGB output. Fog `scene-fog` from 6 to 16 units.

Particles

- Count by tier: 40,000 High, 15,000 Medium, 5,000 Low. One `Points` draw call.
- Size 1.5–3.5 px at pixel ratio 1, attenuated by depth and multiplied by the pixel ratio.
- Colour `scene-particle`; a random 15% use `scene-particle-glow`. A soft disc alpha (smoothstep from 0.5 to 0.15 of the point radius).
- Additive blending in the Lagoon theme, normal blending in the Pearl theme; `depthWrite` off.
- Pointer: particles within 0.18 (normalised screen units) of the pointer are pushed outward with strength 0.35, eased with the `pointer` lerp. No pointer effect on touch devices.
- Thinking pulse: while Pearl streams an answer, particles within radius 3 drift 4% toward the pearl and `scene-particle-glow` rises 20%, easing back over 800 ms.

Post-processing

Tier	Effects
High	Bloom (intensity 0.6, threshold 0.6, mipmap blur), film grain (opacity 0.035), vignette (offset 0.3, darkness 0.6)
Medium	Film grain only
Low and Off	None

Camera

Perspective, field of view 35°, position (0, 0, 8), looking at the origin. The camera does not move with scroll; formations do. A slight pointer parallax (±0.15 units) applies on desktop High and Medium only.

Reduced motion, Low, and Off

- **Reduced motion**: render one frame of the Orbit formation with no pointer effect and no post-processing, then stop the loop.
- **Off** (no WebGL 2, Save-Data, or the footer toggle): the inline SVG poster of the same composition, a pearl drawn from radial gradients with a ring of about 120 small circles in `scene-particle`.
- The poster and the first WebGL frame must match closely enough that the cross-fade reads as the scene waking up, not as a swap.

Pearl chat

Pearl looks like part of the site, not a third-party widget: same type, same lagoon, plain-text answers, and the pearl as its face.

Voice

Pearl is warm, concise, and a little witty, and always talks about Kushwant in the third person. Answers run 2 to 4 sentences, lists at most 5 items, and most end with one useful next step.

Situation	Pearl says
Opening	"Hi, I'm Pearl, Kushwant's AI guide. Ask about his work, check his fit for a role, or scope a project with him."
Known answer	"Yes. His latest is a booking assistant for a Hyderabad clinic: patients book in the website chat, and the doctor confirms from WhatsApp."
Not in the knowledge base	"I don't have that in Kushwant's notes. Want me to pass the question to him?"
Salary or rates	"That's best discussed with Kushwant directly. I can send him your details, or you can book a call."
Off-topic request	"I'm only here to help with questions about Kushwant and his work. Try asking which projects use AI."
Pasted job description	"Here's how Kushwant lines up with this role." followed by the fit card
Rate limited	"You've asked a lot of questions in a few minutes. I can answer again at 8:40 PM; meanwhile, his email is below."
Quick answers mode	"I'm in quick answers mode right now, so my replies are shorter. For anything detailed, message Kushwant."

Parts and their look

- **Text**: plain paragraphs in `body` at 16 px, `pearl`. Markdown allowed: bold, lists, links. No headings inside answers.
- **Source chips**: `Chip` kind source, one per section used, under the text.
- **Project card** and **résumé card**: `reef` -inside- `reef` is not allowed, so chat cards sit on `lagoon` with a 1 px `line` outline, `radius-md`, and one or two sm buttons.
- **Fit card** (JD match, P1): two short lists, "Matches" (each item links to its evidence) and "Gaps" (honest, in `mist`), then actions: Download résumé, Book a call.
- **Confirm card**: everything Pearl will send, word for word, then "Send message" (primary sm) and "Edit" (secondary sm). After sending: "Sent. Kushwant usually replies within a day."

Starters by context

Where the visitor is	Starters
Anywhere, first open	"Who is Kushwant?", "Show me his best project", "Is he available for freelance?", "Check his fit for my role"
Freelance	"What has he built for clients?", "How does a project with him work?", "Help me scope my project"
Projects	"What's the hardest problem he solved?", "Which projects use AI?"
A case-study page	"How was this built?", "What would he do differently?"

Phone sheet

90svh, `radius-lg` top corners, a 36 × 4 px `line-strong` drag handle (swipe down or tap the backdrop to close), composer padded by `env(safe-area-inset-bottom)`. The keyboard pushes the composer up; the log keeps the newest message in view.

Keyboard

Enter sends; Shift+Enter adds a line; Escape closes; Tab moves through messages' links and buttons in order. Ctrl or ⌘ K opens the command palette, which has an "Ask Pearl" entry that passes the typed text.

Never

- Model names, token counts, "As an AI language model", or system prompt details.
- Emoji, exclamation-heavy enthusiasm, or first-person claims as Kushwant.
- Sending anything to Kushwant before the visitor presses Send message on the confirm card.

Writing

Words are part of the interface: plain verbs, sentence case, the visitor's vocabulary, and one job per element.

Rules

- Name things by what visitors recognise: "Résumé", "Freelance", "Ask Pearl", never "CV download module" or "AI agent".
- A control says what happens, and the confirmation repeats it: "Send message" leads to "Message sent".
- Errors say how to fix the problem, in the interface's voice, without apologising.
- Empty and offline states point to the next action.
- No exclamation marks in interface copy. No emoji anywhere.
- Avoid dashes as asides; use full stops or commas.

Interface copy

Where	Copy	Notes
Hero, primary	View my work	Scrolls to Projects
Hero, secondary	Hire me for a project	Scrolls to Freelance
Hero, quiet	Ask Pearl about me	Opens Pearl with the "Who is Kushwant?" starter sent
Nav	Résumé	Downloads the PDF; the file is named `Kushwant-KR-Resume.pdf`
Freelance badge	Freelance: 1 slot open in November	From `availability.freelance` in config
Freelance CTA	Start a project	Opens Contact with "Freelance project" selected
Freelance CTA, quiet	Scope it with Pearl	Opens Pearl in scoping mode
Case-study card links	Live site, Code, Ask Pearl about this	
Contact heading	Let's build something	
Contact summary	Tell me about a role, a project, or an idea. I reply within 24 hours.	
Contact submit	Send message	Busy: "Sending message"
Contact success toast	Message sent / Kushwant replies within 24 hours.	
Contact failure toast	Couldn't send your message / Check your connection and try again, or email directly.	Stays until dismissed
Email field error	Enter an email address like name@example.com.	
Required field error	Enter your name. / Tell me a little about what you need.	One per field
Copy email	Copy email, then "Email copied"	
Local time	It's 8:08 PM in Hyderabad	Updates every minute
Footer	How this site is built	Links to /colophon
Footer toggles	Motion: Auto, Reduced. 3D: Auto, Low, Off.	Segmented controls
404 heading	This page is lost at sea	
404 text	The link may be old, or the page moved. Head back home, or ask Pearl where to find it.	Buttons: "Go home", "Ask Pearl"
Chat placeholder	Ask about Kushwant's work	
Chat footer	AI can make mistakes. Answers come from Kushwant's portfolio content.	Links "Privacy"
Chat retry	Pearl couldn't answer that. Try again.	Button: "Retry"
Placeholder ribbon (preview builds only)	Sample content	On every entry with `placeholder: true`

Numbers, dates, and money

- Dates: "Jul 2026", ranges with an en dash: "Jan – Jun 2026"; "present" in lowercase.
- Time: "8:08 PM" with IST implied; say "in Hyderabad" when it matters.

- Money: "₹40,000", "₹1,50,000"; ranges "₹40,000 – ₹1,00,000"; "Not sure yet" as the last budget option.
- Percentages and counts in digits: "60% of bookings", "1,200 learners".

Implementation

In the codebase, `tokens.json` stays the single source: a small script turns it into Tailwind v4 theme CSS, and components use utilities named after the tokens (`bg-reef` , `text-mist` , `rounded-pill` , `shadow-focus`). Everything on this page was compiled against Tailwind CSS 4.3.3.

Files

Path	What it is
`design/tokens.json`	A copy of this system's `tokens.json` . Update it from the design system; never edit it by hand.
`scripts/tokens.ts`	Generates `src/styles/tokens.css` . Run `pnpm tokens` after updating `design/tokens.json` .
`src/styles/tokens.css`	Generated: theme variables, the Tailwind theme mapping, z-index utilities.
`src/app/globals.css`	Hand-written: imports, font mapping, type utilities, the pearl utility, base styles.
`src/app/fonts.ts`	`next/font` setup for Anek Latin, Anek Telugu, Martian Mono.
`src/lib/motion.ts`	Motion presets for the `motion` package.
`src/scene/colors.ts`	Reads the `scene-*` tokens for Three.js.

The generator

Add `"tokens": "node scripts/tokens.ts"` to `package.json` scripts; Node 22.18 and later run TypeScript files directly.

```
// scripts/tokens.ts: turns design/tokens.json (exported from the Nacre design system) into src/styles/tokens.css.
// Run with `node scripts/tokens.ts` (Node 22.18+ runs TypeScript directly) or via `pnpm tokens`.
import { readFileSync, writeFileSync } from 'node:fs';

type Value = string | Record<string, string>;
interface Token { name: string; value: Value; usage?: string }
interface Family { tokens: Token[] }

const src = JSON.parse(readFileSync('design/tokens.json', 'utf8'));
const themes: { id: string }[] = src.color.themes;
const first = themes[0].id;

const pick = (t: Token, theme: string): string =>
  typeof t.value === 'string' ? t.value : (t.value[theme] ?? t.value[first]);
// "{lagoon}" is an alias of another token
const css = (v: string): string => v.replace(/^\{([A-Za-z0-9_.-]+)\}$/, 'var(--$1)');
const fam = (key: string): Token[] => ((src[key] as Family | undefined)?.tokens ?? []);

const colors = fam('color');
const shadows = fam('shadow');
const themed = (theme: string, onlyChanged: boolean): string =>
  [
    ...colors.map((t) => [t.name, t] as const),
    ...shadows.map((t) => [`nacre-${t.name}`, t] as const),
  ]
    .filter(([, t]) => !onlyChanged || (typeof t.value !== 'string' && pick(t, theme) !== pick(t, first)))
    .map(([name, t]) => `  --${name}: ${css(pick(t, theme))};`)
    .join('\n');

const plain = (keys: string[]): string =>
  keys.flatMap(fam).map((t) => `  --${t.name}: ${pick(t, first)};`).join('\n');

const out = [
  '/* Generated from design/tokens.json by scripts/tokens.ts. Edit the design system, not this file. */',
  '',
  `:root,\n[data-theme="${first}"] {\n  color-scheme: ${first};\n${themed(first, false)}\n}`,
  ...themes.slice(1).map((t) => `[data-theme="${t.id}"] {\n  color-scheme: ${t.id};\n${themed(t.id, true)}\n}`),
  '',
  `:root {\n${plain(['spacing', 'size', 'zIndex'])}\n}`,
  '',
  '@theme {',
  '  --radius-*: initial;',
  fam('radius').map((t) => `  --${t.name}: ${pick(t, first)};`).join('\n'),
  '}',
  '',
  '@theme inline {',
  colors.map((t) => `  --color-${t.name}: var(--${t.name});`).join('\n'),
  ...shadows.map((t) =>
    t.name === 'shadow-lustre'
      ? '  --inset-shadow-lustre: var(--nacre-shadow-lustre);'
      : `  --${t.name}: var(--nacre-${t.name});`,
  ),
  '}',
  '',
  ...fam('zIndex').map((t) => `@utility ${t.name} {\n  z-index: var(--${t.name});\n}`),
  '',
].join('\n');
```

```
writeFileSync('src/styles/tokens.css', out);
console.log(`src/styles/tokens.css: ${colors.length} colours, ${themes.length} themes`);
```

What it writes (abridged):

```
/* Generated from design/tokens.json by scripts/tokens.ts. Edit the design system, not this file. */
:root,
[data-theme="dark"] {
  color-scheme: dark;
  --lagoon: #0d2627;
  --reef: #123131;
  --pearl: #f3eee6;
  --rose: #f4a7b9;
  /* … every colour token … */
  --scene-fog: var(--lagoon);
  --nacre-shadow-focus: 0 0 0 2px #0d2627, 0 0 0 4px #f4a7b9;
}
[data-theme="light"] {
  color-scheme: light;
  --lagoon: #f0f3f1;
  --pearl: #0d2627;
  --rose: #a83d5a;
  /* … only the values that change … */
}
@theme {
  --radius-*: initial;
  --radius-xs: 4px;
  --radius-sm: 8px;
  --radius-md: 14px;
  --radius-lg: 22px;
  --radius-pill: 999px;
}
@theme inline {
  --color-lagoon: var(--lagoon);
  --color-rose: var(--rose);
  /* … one --color-* per colour token … */
  --inset-shadow-lustre: var(--nacre-shadow-lustre);
  --shadow-float: var(--nacre-shadow-float);
  --shadow-focus: var(--nacre-shadow-focus);
}
@utility z-nav {
  z-index: var(--z-nav);
}
```

Shadows get a `nacre-` prefix on their raw variables so Tailwind's `--shadow-*` names can point at them without referring to themselves. `shadow-lustre` becomes Tailwind's `inset-shadow-lustre`, so it combines with `shadow-float` or `shadow-focus` on one element.

globals.css

```
@import "tailwindcss";
@import "../styles/tokens.css";

@theme inline {
  --font-sans: var(--font-anek), var(--font-anek-telugu), ui-sans-serif, system-ui, sans-serif;
  --font-telugu: var(--font-anek-telugu), var(--font-anek), sans-serif;
  --font-mono: var(--font-martian), ui-monospace, "SF Mono", Menlo, monospace;
}

@theme {
  --ease-nacre-out: cubic-bezier(0.22, 1, 0.36, 1);
  --ease-nacre-in-out: cubic-bezier(0.65, 0, 0.35, 1);
  --animate-shimmer: shimmer 1600ms ease-in-out infinite;
  @keyframes shimmer {
    0%, 100% { filter: hue-rotate(0deg) brightness(1); }
    50% { filter: hue-rotate(-24deg) brightness(1.1); }
  }
}

/* One class per type style: size, leading, weight, tracking and width together */
@utility type-display { font-size: clamp(3rem, 2.5rem + 2.222vw, 4.5rem); line-height: 1.02; font-weight: 760; letter-spacing: -0.015em;
font-variation-settings: "wdth" 114; text-wrap: balance; }
@utility type-h1 { font-size: clamp(3rem, 2.75rem + 1.111vw, 3.75rem); line-height: 1.05; font-weight: 720; letter-spacing: -0.01em;
font-variation-settings: "wdth" 110; text-wrap: balance; }
@utility type-h2 { font-size: clamp(2.25rem, 2rem + 1.111vw, 3rem); line-height: 1.08; font-weight: 700; letter-spacing: -0.01em; font-
variation-settings: "wdth" 108; text-wrap: balance; }
@utility type-h3 { font-size: clamp(1.5rem, 1.25rem + 1.111vw, 2.25rem); line-height: 1.15; font-weight: 660; letter-spacing: -0.005em;
font-variation-settings: "wdth" 104; text-wrap: balance; }
@utility type-h4 { font-size: clamp(1.3125rem, 1.25rem + 0.2778vw, 1.5rem); line-height: 1.25; font-weight: 620; }
@utility type-lead { font-size: clamp(1.125rem, 1.0625rem + 0.2778vw, 1.3125rem); line-height: 1.5; font-weight: 460; }
@utility type-body { font-size: clamp(1rem, 0.9583rem + 0.1852vw, 1.125rem); line-height: 1.6; font-weight: 460; }
@utility type-ui { font-size: 1rem; line-height: 1.25; font-weight: 600; }
@utility type-small { font-size: 0.875rem; line-height: 1.5; font-weight: 520; }
@utility type-caption { font-size: 0.75rem; line-height: 1.4; font-weight: 540; }
@utility type-code { font-family: var(--font-mono); font-size: 0.9375rem; line-height: 1.6; font-weight: 420; font-variation-settings:
"wdth" 87.5; }

/* The pearl: the only gradient in the system */
```

```
@utility pearl {
  border-radius: 9999px;
  background:
    radial-gradient(circle at 34% 28%, #fff 0 9%, #fff0 32%),
    radial-gradient(circle at 72% 78%, var(--scene-sheen-aqua) 0 6%, #9ee6e000 48%),
    radial-gradient(circle at 22% 74%, var(--scene-sheen-rose) 0 5%, #f4a7b900 46%),
    radial-gradient(circle at 50% 46%, var(--scene-pearl) 0 55%, #b7c2c6 100%);
  box-shadow: inset -0.05em -0.07em 0.12em #0d262766;
}

@layer base {
  html { background: var(--lagoon); }
  body { background: var(--lagoon); color: var(--pearl); font-family: var(--font-sans); font-weight: 460; -webkit-font-smoothing:
antialiased; }
  ::selection { background: var(--rose); color: var(--on-rose); }
  :focus-visible { outline: 2px solid var(--rose); outline-offset: 2px; }
  section[id] { scroll-margin-top: var(--nav-height); }
  @media (prefers-reduced-motion: reduce) {
    *, *::before, *::after { animation-duration: 0.01ms !important; animation-iteration-count: 1 !important; transition-duration: 0.01ms
!important; scroll-behavior: auto !important; }
  }
}
```

Components that draw their own focus ring add `focus-visible:outline-none focus-visible:shadow-focus`; everything else gets the base outline, so nothing is ever left without one.

Fonts

```
// src/app/fonts.ts
import { Anek_Latin, Anek_Telugu, Martian_Mono } from 'next/font/google';

export const anek = Anek_Latin({ subsets: ['latin', 'latin-ext'], axes: ['wdth'], variable: '--font-anek', display: 'swap' });
export const anekTelugu = Anek_Telugu({ subsets: ['telugu'], axes: ['wdth'], variable: '--font-anek-telugu', display: 'swap', preload:
false });
export const martian = Martian_Mono({ subsets: ['latin'], axes: ['wdth'], variable: '--font-martian', display: 'swap', preload: false
});
```

In `src/app/layout.tsx`: `<html lang="en-IN" data-theme="dark" className={` ${anek.variable} ${anekTelugu.variable} ${martian.variable} `}>`. Only Anek Latin preloads; Telugu and mono load when a page first uses them. All three are under the SIL Open Font License.

Which utility for which token

Token family	Utilities
Colour	`bg-lagoon`, `bg-reef`, `bg-shoal`, `text-pearl`, `text-mist`, `text-rose`, `bg-rose`, `text-on-rose`, `border-line`, `border-line-strong`, `ring-line`, `bg-seaglass-soft` `text-seaglass`, … one per colour token
Radius	`rounded-xs`, `rounded-sm`, `rounded-md`, `rounded-lg`, `rounded-pill` (Tailwind's default radii are cleared)
Shadow	`inset-shadow-lustre`, `shadow-float`, `shadow-focus`
Spacing	Tailwind's 4 px scale: `space-1` = 1, `space-2` = 2, `space-3` = 3, `space-4` = 4, `space-5` = 6, `space-6` = 8, `space-7` = 12, `space-8` = 16, `space-9` = 24, `space-10` = 32 (as in `p-6`, `gap-12`)
Type	`type-display` … `type-caption`, `type-code`, `font-telugu`
Size	`max-w-(--container-max)`, `max-w-(--measure)`, `h-(--nav-height)`, `size-(--launcher-size)`
Z-index	`z-canvas`, `z-content`, `z-nav`, `z-launcher`, `z-chat`, `z-palette`, `z-toast`, `z-skip`
Motion	`ease-nacre-out`, `ease-nacre-in-out`, `animate-shimmer`
The pearl	`pearl` with a size, as in `pearl size-[0.32em]`

Three.js colours

```
// src/scene/colors.ts: read once on mount and again when data-theme changes
import { Color } from 'three';

const read = (name: string) =>
  getComputedStyle(document.documentElement).getPropertyValue(`--${name}`).trim();

export function sceneColors() {
  return {
    particle: new Color(read('scene-particle')),
    glow: new Color(read('scene-particle-glow')),
    pearl: new Color(read('scene-pearl')),
    sheenRose: new Color(read('scene-sheen-rose')),
    sheenAqua: new Color(read('scene-sheen-aqua')),
    fog: new Color(read('scene-fog')),
  };
}
```

`three` converts these sRGB hex values into its linear working space, so the canvas matches the CSS.

Motion presets

```
// src/lib/motion.ts
export const ease = { out: [0.22, 1, 0.36, 1], inOut: [0.65, 0, 0.35, 1] } as const;
export const duration = { press: 0.1, quick: 0.16, base: 0.24, panel: 0.32, scene: 1.2 } as const;
export const spring = { panel: { type: 'spring', stiffness: 380, damping: 34 } } as const;
```

Wrap the app in `<MotionConfig reducedMotion="user">` so every Motion animation honours the visitor's setting.

Guardrails for people and coding agents

- Tailwind's default palette still compiles (`text-rose-400` builds without a warning), so CI fails on it: `grep -rEn "\b(bg|text|border|ring|fill|stroke|outline|decoration)-((slate|gray|zinc|neutral|stone|red|orange|amber|yellow|lime|green|emerald|teal|cyan|sky|blue|indigo|violet|purple|f[0-9]{2,3}|black|white)\b" src/` must find nothing.
- No hex colours in components: only `src/styles/tokens.css`, the `pearl` utility, and `src/scene/` may contain them.
- No arbitrary font sizes (`text-[17px]`); use a `type-*` utility.

If you add shadcn/ui

Point its variables at Nacre tokens instead of keeping its defaults:

```
:root {
  --background: var(--lagoon); --foreground: var(--pearl);
  --card: var(--reef); --card-foreground: var(--pearl);
  --popover: var(--shoal); --popover-foreground: var(--pearl);
  --primary: var(--rose); --primary-foreground: var(--on-rose);
  --secondary: var(--shoal); --secondary-foreground: var(--pearl);
  --muted: var(--reef); --muted-foreground: var(--mist);
  --accent: var(--rose-soft); --accent-foreground: var(--rose);
  --destructive: var(--coral);
  --border: var(--line); --input: var(--line-strong); --ring: var(--rose);
}
```

Fonts and licences

The three typefaces ship as variable WOFF2 files in `fonts/`, each under the SIL Open Font License 1.1 (licence texts in `fonts/licenses/`).

Family	File	Axes	Role
Anek Latin (Ek Type)	`AnekLatin-Variable.woff2`	Weight 100 to 800, width 75 to 125	Every heading and all running text
Anek Telugu (Ek Type)	`AnekTelugu-Variable.woff2`	Weight 100 to 800, width 75 to 125	Telugu text, through the font stack
Martian Mono (Evil Martians)	`MartianMono-Variable.woff2`	Weight 100 to 800, width 75 to 112.5	Code only